

\* Some non-local holomorphic correspondence between spacetimes and some  $\mathbb{C}$  projective variety

$$\leadsto \mathbb{C} \text{ 4D } M^4 : M^4_{\mathbb{C}}$$

$$(x^0, x^1, x^2, x^3) \in \mathbb{C}^4 \mid ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

$$\text{Lorentzian } (1,3): x^a \in \mathbb{R}$$

$$\text{Euclidean } \mathbb{R}^4: x^0 \in \mathbb{R}, x^1, x^2, x^3 \in i\mathbb{R}$$

$$SO(4, \mathbb{C}) \cong SL(2, \mathbb{C}) \times SL(2, \mathbb{C}) / \mathbb{Z}_2$$

$$\left( \frac{1}{2}, 0 \right) \oplus \left( 0, \frac{1}{2} \right)$$

$$A^\mu : \left( \frac{1}{2}, \frac{1}{2} \right) : V = V^\mu \partial_\mu \in TM^4$$

$$V^a \mapsto \left( \sigma_a^{\alpha\dot{\beta}} / \sqrt{2} \right) V^a = \frac{1}{\sqrt{2}} \begin{pmatrix} V^0 + V^1 & V^2 + iV^3 \\ V^2 - iV^3 & V^0 - V^1 \end{pmatrix} \equiv V^{\alpha\dot{\beta}}$$

$$\epsilon_{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \epsilon_{\dot{\alpha}\dot{\beta}}$$

$$V^\beta \epsilon_{\beta\alpha} = V_\alpha, \quad V^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}} V_{\dot{\beta}}$$

$$\eta(V, W) = \eta_{\mu\nu} V^\mu V^\nu = \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} V^{\alpha\dot{\alpha}} V^{\beta\dot{\beta}}$$

\* Unconstrained way of describing null-vectors

$$ds^2 = \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} dx^{\alpha\dot{\alpha}} dx^{\beta\dot{\beta}}$$

$$V^2 = \epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} V^{\alpha\dot{\alpha}} V^{\beta\dot{\beta}} = 2 \det(V^{\alpha\dot{\alpha}}) = 0 \quad (\text{Null})$$

$$V^{\alpha\dot{\alpha}}_{null} = \lambda^\alpha \lambda^{\dot{\alpha}} \iff \det(V^{\alpha\dot{\alpha}}) = 0$$

Defined up to  $U(1)$ :  $\lambda^\alpha \rightarrow r \lambda^\alpha \quad r \in \mathbb{C}^*$

$$\lambda^{\dot{\alpha}} \rightarrow \frac{1}{r} \lambda^{\dot{\alpha}}$$

$\mathbb{C}^*$ : Little group

$$SO(2) \cong U(1)$$

$$\langle VW \rangle = \epsilon_{\alpha\beta} V^\alpha V^\beta$$

$$[VW] = \epsilon_{\dot{\alpha}\dot{\beta}} V^{\dot{\alpha}} V^{\dot{\beta}}$$

Twistor Space:  $\mathbb{CP}^3$  define homogeneous coords

$$Z^A = (z^0, z^1, z^2, z^3) \text{ s.t. } Z^A \neq 0$$

$$\text{and } (Z)^\lambda \sim r (Z^\lambda) \quad r \in \mathbb{C}^*$$

$$[Z] = \left( 1, \frac{z^1}{z^0}, \frac{z^2}{z^0}, \frac{z^3}{z^0} \right) \quad (\text{Affine Coords})$$

$$Z^A = (\mu^{\dot{0}}, \lambda_\alpha)$$

$$\mu^{\dot{1}} = z^0 \quad \lambda_1 = z^2$$

$$\mu^{\dot{2}} = z^1 \quad \lambda_2 = z^3$$

\* Have a relation between  $\mathbb{CP}^3$  and  $M^4$

via incidence relations

$$\mu^i = x^{\alpha i} \lambda_{\alpha}$$

$$\begin{pmatrix} \mu^1 \\ \mu^2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} x^0 + x^1 & x^2 + ix^3 \\ x^2 + ix^3 & x^0 - x^1 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}$$

$$= \begin{pmatrix} (x^0 + x^1) \lambda_1 + (x^2 + ix^3) \lambda_2 \\ (x^2 + ix^3) \lambda_1 + (x^0 - x^1) \lambda_2 \end{pmatrix}$$

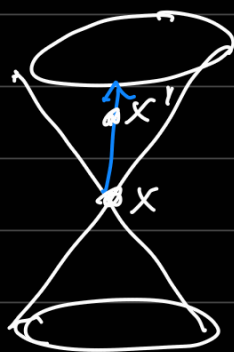
$$\mu^1 = (x^0 + x^1) \lambda_1 + (x^2 + ix^3) \lambda_2$$

$$\mu^2 = (x^2 + ix^3) \lambda_1 + (x^0 - x^1) \lambda_2$$

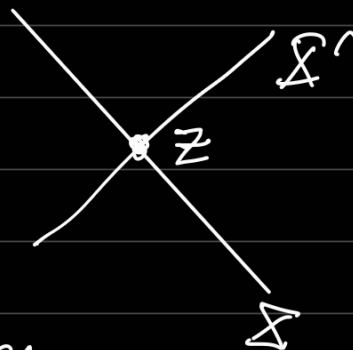
• Not every point in  $\mathbb{CP}^3 \longleftrightarrow$  Finite point in  $\mathbb{M}^4$   
 • Complex line in  $\mathbb{CP}^3$ : linearly embedded Riemann sphere

$$\mathbb{Z}^4 = (\mu^i, \lambda_{\alpha} = 0)$$

• Points in  $\mathbb{M}^4 \longleftrightarrow$  extended geometric objects in  $\mathbb{PT}$



holomorphic  
(non-local)



$$\mu^{\alpha} = x^{\alpha i} (0) = 0 \leftarrow \text{can't happen}$$

Since  $\mathbb{Z}^I$  are homogeneous:

$\rightarrow$  Remove subset from  $\mathbb{CP}^3$

$$\mathbb{PT} = \{ \mathbb{Z}^I \in \mathbb{CP}^3 \mid \lambda_{\alpha} \neq 0 \}$$

$$\mathbb{PT} \subset \mathbb{CP}^3$$

Twistor correspondence: given  $x \in \mathbb{M}$ , what do we get  
 $(\mathbb{M} \rightarrow \mathbb{PT})$  in  $\mathbb{PT}$ ?

$$\mu^i = x^{\alpha i} \lambda_{\alpha} \quad x^{\alpha i}: \text{fixed}$$

$$\hookrightarrow \mathbb{CP}^2 / \mathbb{CP}^1 \subset \mathbb{CP}^3 / \mathbb{CP}^1 \Rightarrow \mathbb{CP}^1 \subset \mathbb{CP}^3$$

point  $x \in \mathbb{M}^4$

$\mathbb{PT}$

$\hookrightarrow$  Holomorphic, linearly embedded  $\mathbb{CP}^1 \subset \mathbb{PT}$

a "Twistor-line"  $X \cong \mathbb{CP}^1 \subset \mathbb{PT}$

\* Take a fixed pt  $Z^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha}) \in \mathbb{PT}$

$$\mathbb{PT} \rightarrow \mathbb{M}^{\mathbb{C}}$$

Take  $Z^A = X \wedge Y \Rightarrow \mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$   
 $\mu^{\dot{\alpha}} = y^{\alpha\dot{\alpha}} \lambda_{\alpha}$

$$0 = (x-y)^{\alpha\dot{\alpha}} \lambda_{\alpha}$$

$$(x-y)^{\alpha\dot{\alpha}} \lambda_{\alpha} \Rightarrow (x-y)^{\alpha\dot{\alpha}} = \lambda^{\alpha} \lambda^{\dot{\alpha}} e^{-1}$$

Since

$$(x-y)^{\alpha\dot{\alpha}} \lambda^{\beta} \epsilon_{\beta\alpha} = 0$$

Contracting two spinors  $\chi^{\alpha} \lambda_{\alpha} = \chi^{\alpha} \epsilon_{\beta\alpha} \lambda^{\beta} = 0$

means either  $\chi^{\alpha} = 0$  or  $\chi^{\alpha} = r \lambda^{\alpha}$

Since  $x=y$  is trivial this means

$$(x-y)^{\alpha\dot{\alpha}} = e^{\alpha\dot{\alpha}} \lambda^{\alpha} \lambda^{\dot{\alpha}}$$

$\Rightarrow x, y$  are null-separated in  $\mathbb{M}^{\mathbb{C}}$

$(\overline{X}, \overline{Y})$  : intersection  $\mathbb{PT}$  iff  $x, y \in \mathbb{M}^{\mathbb{C}}$  are null separated

$$(x-y)^{\alpha\dot{\alpha}} = \lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}} : \text{Parameterized by } \tilde{\lambda}^{\dot{\alpha}} \neq 0$$

\* So, a point  $Z$  in  $\mathbb{PT} \longleftrightarrow$  totally null  $\mathbb{CP}^2 \subset \mathbb{M}$  parameterized by the choice  $\tilde{\lambda}^{\dot{\alpha}}$ .

Zero Rest Mass fields on  $\mathbb{M}^{\mathbb{C}}$ : (4D)  $s \geq 0$

(4) decomposed into helicity  $\pm s$  parts

$$+s: \tilde{\phi}_{\dot{\alpha}_1 \dots \dot{\alpha}_{2s}}(x)$$

$$-s: \phi_{\alpha_1 \dots \alpha_{2s}}(x)$$

$$* \underline{s=1} \quad A_{\alpha\dot{\alpha}}, \quad F_{\alpha\dot{\alpha}}{}^{\beta\dot{\beta}} = \partial_{\alpha\dot{\alpha}} A^{\beta\dot{\beta}} - \partial^{\beta\dot{\beta}} A_{\alpha\dot{\alpha}}$$

$$F_{\alpha\dot{\alpha}}{}^{\beta\dot{\beta}} = -F_{\beta\dot{\beta}}{}^{\alpha\dot{\alpha}}$$

$$(i) (\text{Anti sym } \dot{\alpha}\dot{\beta}) \hat{=} (\text{Sym } \alpha\beta)$$

$$(ii) (\text{ " " } \alpha\beta) \hat{=} (\text{Sym } \dot{\alpha}\dot{\beta})$$

So can decompose  $F$  into  $\text{sym} \oplus \Lambda^2$

$$F_{\alpha\dot{\alpha}}{}^{\beta\dot{\beta}} = \underbrace{\epsilon_{\alpha\beta} \tilde{F}_{\dot{\alpha}\dot{\beta}}}_{\text{II}} + \underbrace{\epsilon_{\dot{\alpha}\dot{\beta}} \hat{F}_{\alpha\beta}}_{\text{II}}$$

$$\tilde{\phi}_{\dot{\alpha}\dot{\beta}}$$

$$\phi_{\alpha\beta}$$

$F$  decomposes into 2 helicity states  $\pm 1$  (Anti)

decomposes into its helicity states. Dual/HK self dual.

## \* Zero Rest Mass equations

$$\partial^{\alpha\dot{\alpha}} \tilde{\phi}_{\dot{\alpha}_1 \dots \dot{\alpha}_{2s}} = 0 \quad : \text{Bianchi for } s=1$$

$$\partial^{\alpha\dot{\alpha}} \phi_{\alpha_1 \dots \alpha_{2s}} = 0$$

$s=0$ : wave equation

ex: Momentum eigenstates: states  $\propto$  to  $e^{ik \cdot x}$  for

$$K^\mu K_\mu = 0$$

$$\hookrightarrow K^{\alpha\dot{\alpha}} = K^\alpha \tilde{K}^{\dot{\alpha}}$$

$$\begin{cases} \tilde{\phi}_{\dot{\alpha}_1 \dots \dot{\alpha}_{2s}} = \tilde{K}_{\dot{\alpha}_1} \dots \tilde{K}_{\dot{\alpha}_{2s}} e^{ik \cdot x} \\ \phi_{\alpha_1 \dots \alpha_{2s}} = K_{\alpha_1} \dots K_{\alpha_{2s}} e^{ik \cdot x} \end{cases}$$

Penrose Transform: {Helicity  $h$ , zero rest mass fields on  $\mathbb{M}^4$ }

$$\cong H^1(\mathbb{P}^1, \mathcal{O}(2h-2))$$

$\mathcal{O}(h)$ : sheaf of locally homogeneous  $\mathbb{P}^1$ s which are homogeneous of weight  $h$  on  $\mathbb{P}^1$ .

i.e.  $f(rz) = r^h f(z)$

\* Suppose we're given  $f \in \Omega^{0,1}(\mathbb{P}^1, \mathcal{O}(2h-2))$   $h \geq 0$   
for  $f \sim f_{\bar{z}} d\bar{z}$

Can build a spacetime field:  $\phi_{\alpha_1 \dots \alpha_{2s}}(x)$

$$\phi_{\alpha_1 \dots \alpha_{2s}}(x) = \int_{x \in \mathbb{CP}^1} D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2s}} f|_{\mathbb{X}} \quad \leftarrow \text{Pull back}$$

$$D\lambda \equiv \lambda^\alpha d\lambda_\alpha \in \Omega^{1,0}(\mathbb{P}^1, \mathcal{O}(2))$$

$$\text{since } D(r\lambda) = r\lambda^\alpha d(r\lambda_\alpha) = r^2 \lambda^\alpha d\lambda_\alpha$$

$$f|_{\mathbb{X}} = f(\mu^{\dot{\alpha}} = x^{\dot{\alpha}\alpha} \lambda_\alpha, \lambda_\alpha)$$

Restricting to  $\mathbb{X}$  means restricting to the incidence relation.

weight of  $D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2s}}$

$$(2 + 2h) \quad \leftarrow$$

$$f|_{\mathbb{X}} : 2h - 2 \quad \text{cancels}$$

\* Provided  $\partial f = 0$ : we can compute

$$\begin{aligned} \partial^{\alpha\bar{\alpha}} \phi_{\alpha_1 \dots \alpha_{2h+1}} &= \int_{\Sigma = \mathbb{CP}^1} \partial^{\alpha\bar{\alpha}} (D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h+1}} f|_{\Sigma}) \\ &= \int D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h+1}} \frac{\partial}{\partial x_{\alpha\bar{\alpha}}} (f|_{\Sigma}) \\ &= \int D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h+1}} \lambda^{\alpha\bar{\alpha}} \frac{\partial}{\partial u_{\bar{\alpha}}} (f) \\ &= 0 \end{aligned}$$

So long as  $f$  is holomorphic  $\left( \partial_{\alpha\bar{\alpha}} = \lambda_{\alpha} \frac{\partial}{\partial u_{\bar{\alpha}}} \right)$   
 $\bar{\partial}^2 = 0$  so we can finally get  
 $\bar{\partial} f = 0$ : "exact":  $f = \bar{\partial} g$ ,  $g \in \Omega^0(\mathbb{P}^1, \mathcal{O}(2h-2))$

$$\phi_{\alpha_1 \dots \alpha_{2h}} = \int_{\Sigma} D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h}} \bar{\partial} g|_{\Sigma} = \int \bar{\partial} (D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h}} g) = 0$$

(Trivial field) To get nontrivial sol<sup>n</sup>:

$$\{ f | \bar{\partial} f = 0 \} / \{ f | f = \bar{\partial} g \} := H^0(\mathbb{P}^1, \mathcal{O}(2h-2))$$

For  $h > 0$ :

$$\Phi_{\alpha_1 \dots \alpha_{2h}}(x) = \int_{\Sigma} D\lambda \wedge \frac{\partial}{\partial u^{\alpha_1}} \dots \frac{\partial}{\partial u^{\alpha_{2h}}} f|_{\Sigma}$$



Momentum Eigenstates:  $\phi_{\alpha_1 \dots \alpha_{2h+1}} \approx k_{\alpha_1} \dots k_{\alpha_{2h+1}} e^{ik \cdot x}$

Q: What is the Penrose expression? i.e. to what

cohomology does this exist in?

Wick-delta:  $\bar{\partial}(z) = \delta(x) \delta(y) d\bar{z} = \frac{1}{2\pi i} d\bar{z} \frac{\partial}{\partial \bar{z}} \left( \frac{1}{z} \right)$   
 $= \frac{1}{2\pi i} d\bar{z} \bar{\partial} \left( \frac{1}{z} \right)$

$$\int_{\mathbb{C}} f(z) dz \wedge \bar{\partial}(z) = f(0)$$

\* Twistor rep:

$$f^{\alpha}(z) = \left( \frac{\langle \xi \lambda \rangle}{\langle \xi \mu \rangle} \right)^{2h-1} \bar{\delta}(\langle \lambda \mu \rangle) \exp \left[ \frac{\mu \tilde{\lambda}}{\langle \xi \lambda \rangle} \right]$$

\* Have a limit of fixed momentum  $\langle \lambda \mu \rangle$ .

$\xi^{\alpha}$ : Arbitrary fixed spinor (SL(2, C) invariant)

"Gauge Artifact"

exp: weight = 0,  $(2h-1) : \lambda$

weight  $f$  :  $2h-2$

$$f^{(k)} \in \mathcal{O}(2h-2) \otimes \Omega^{0,1}(\mathbb{P}^1)$$

$\Rightarrow$  Since  $\bar{\partial} f^{(k)} = 0$  :  $f^{(k)}$  holomorphic on  $\mathbb{P}^1$

and  $f^{(k)}$  is not exact

$$f^{(k)} \in H^{0,1}(\mathbb{P}^1, \mathcal{O}(2h-2))$$

$\Rightarrow$  By the Penrose Transform we should have a zero rest mass field of helicity  $h$

take negative helicity  $h < 0$ : Restrict  $f$  to a Twistor

line  $\Sigma$  : ( $\mu^{\dot{\alpha}} = x^{\alpha\dot{\beta}} \lambda_{\beta}$ )

$$f|_{\Sigma} = \left( \frac{\langle \xi \lambda \rangle}{\langle \xi \kappa \rangle} \right)^{2h-1} \delta(\langle \lambda \kappa \rangle) \exp \left[ \frac{i \langle \lambda | x | \tilde{\kappa} \rangle \langle \xi \kappa \rangle}{\langle \xi \lambda \rangle} \right]$$

( $[\mu] = \langle \lambda | x$ )

$\therefore$

$$\begin{aligned} \phi_{\alpha_1 \dots \alpha_{2h}} &= \int D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h}} f|_{\Sigma} \\ &= \int D\lambda \wedge \lambda_{\alpha_1} \dots \lambda_{\alpha_{2h}} \left( \frac{\langle \xi \lambda \rangle}{\langle \xi \kappa \rangle} \right)^{2h-1} \delta(\langle \lambda \kappa \rangle) \exp \left[ \frac{i \langle \lambda | x | \tilde{\kappa} \rangle \langle \xi \kappa \rangle}{\langle \xi \lambda \rangle} \right] \end{aligned}$$

\* Integral only has support at  $\langle \lambda \kappa \rangle = 0$

i.e.  $\lambda^{\alpha} = r \kappa^{\alpha}$

$i \langle \kappa | x | \tilde{\kappa} \rangle$

$$\therefore \phi_{\alpha_1 \dots \alpha_{2h}} = \kappa_{\alpha_1} \dots \kappa_{\alpha_{2h}} e^{i \langle \kappa | x | \tilde{\kappa} \rangle}$$

$$\phi_{\alpha_1 \dots \alpha_{2h}} = \kappa_{\alpha_1} \dots \kappa_{\alpha_{2h}} e^{i \langle \kappa | x$$

(on shift)



(set  $\kappa_i = p_i$ )

$$\Rightarrow \phi_{\alpha_1 \dots \alpha_{2h}} = p_{\alpha_1} \dots p_{\alpha_{2h}} e^{i p x}$$

\* Extend  $\delta(z)$  for Super symmetric theories

• For positive helicity  $h \geq 0$

$$\hat{\phi}_{\dot{\alpha}_1 \dots \dot{\alpha}_{2h}}(x) = \int_{\Sigma = \mathbb{CP}^1} D\lambda \wedge \frac{\partial}{\partial \mu^{\dot{\alpha}_1}} \dots \frac{\partial}{\partial \mu^{\dot{\alpha}_{2h}}} i f|_{\Sigma}$$

$$\begin{aligned} \frac{\partial}{\partial \mu^{\dot{\alpha}}} f|_{\Sigma} &= \left[ \left( \frac{\langle \xi \lambda \rangle}{\langle \xi \kappa \rangle} \right)^{2h-1} \delta(\langle \lambda \kappa \rangle) \right] \frac{\partial}{\partial \mu^{\dot{\alpha}}} ([\mu \tilde{\kappa}]) e^{i(\dots)} \\ &= [\dots] \tilde{\kappa}_{\dot{\alpha}} e^{i(\dots)} \end{aligned}$$

$$\therefore \frac{\partial}{\partial \mu^{\dot{\alpha}_1}} \dots \frac{\partial}{\partial \mu^{\dot{\alpha}_{2h}}} f|_{\Sigma} = [\dots] \kappa_{\dot{\alpha}_1} \dots \kappa_{\dot{\alpha}_{2h}} e^{i(\dots)}$$

$\Rightarrow$  Again Supp only on  $\langle \lambda \kappa \rangle = 0$ :

$i \langle \kappa | x | \tilde{\kappa} \rangle$

( $\kappa \leftrightarrow p$ )

$$\Phi_{\alpha_1 \dots \alpha_{2n}}(x) = K_{\alpha_1} \dots K_{\alpha_{2n}} \epsilon$$

Moral of the Story:

$$\text{Holomorphicity on } \mathbb{P}T \leftrightarrow \text{E.O.M on } M^D$$

$$\bar{\partial} f = 0 \quad \longleftrightarrow \quad (\text{Z.F.M}) \text{ equation}$$

Ward Correspondence:  $\rightarrow$  Flat along  $\Sigma$  (any surface line)

Holomorphic Vect Bundles on  $\mathbb{P}T$

$\updownarrow$

Self-Dual Yang Mills on  $M^D$

\* Non linear Granten:

(integrable deformations of  $\mathbb{C}$ -structure on  $\mathbb{P}T$ )

$\updownarrow$

(Self-dual Conformal 4 Manifolds)

$\rightarrow$  Holomorphic Vector Bundle: Holomorphic  $T\mathbb{P}T$

So Granten is a special case of Ward.

Amplitudes of tree-level scattering in Pure

Yang-Mills:

Work  $\left\{ \begin{array}{l} \text{on shell, polarization, \& } \\ \text{color algebra.} \end{array} \right\}$



Feynman

Gluon:  $K^{\alpha\dot{\alpha}} = K^{\alpha} \bar{K}^{\dot{\alpha}}$

Can Lorentz boost s.t. Gluons have  $\downarrow$

helicity  $\pm 1$

$$(C_s + C_t + C_u) = 0$$

Color factors:  $C_s \equiv f^{a_1 a_2 b} f^{b a_3 a_4}$ ,  $C_t, C_u$

$$M_n^{col}(1^{a_1}, \dots, n^{a_n}) = \sum_{\sigma \in S_n / \mathbb{Z}_n} \text{Tr}(T^{a_{\sigma(1)}} \dots T^{a_{\sigma(n)}}) A_n^{(0)}(\sigma(1), \dots, \sigma(n))$$

$G = SU(N)$ ,  $A^\mu$  is in the Adjoint

(all the physics is here)

$$\text{i.e. } A^\mu = \begin{pmatrix} A_1^\mu \\ \vdots \\ A_{N^2-1}^\mu \end{pmatrix} \quad A_\mu = A_\mu^a T^a$$

$$a = 1, \dots, N^2 - 1$$

\*  $A_\mu^{col} \equiv A_\mu$ . For Pure Yang Mills we have

$$\mathcal{L} = -\frac{1}{4} \text{Tr}(F_{\mu\nu} F^{\mu\nu})$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - g[A_\mu, A_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - g^2 A_\mu A_\nu$$

$$= \text{Tr}(-\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu - i f_{abc} g \partial^\mu A^\nu A^\mu A_\nu + \frac{1}{4} A^\mu A^\nu A_\mu A_\nu)$$

ex: Do 4-Gluon Scattering



Work in Gervais-Neveu gauge

$$H_{int} = \partial_\mu A_\nu - \frac{ig}{\sqrt{2}} A_\mu A_\nu$$

$$\langle A^\mu A^\nu A^\lambda A^\sigma \rangle \sim \langle A^\mu A^\nu A^\lambda A^\sigma \left( 1 - i \int d^4x H_{int} + \dots \right) \rangle$$

lowest order:

$$A_\mu \sim g^2 (A_\mu [1234] \text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4}) + \text{perm of } (234))$$

$$\langle 0 | T \{ A^\mu(x_1) A^\nu(x_2) A^\lambda(x_3) A^\sigma(x_4) (-i \frac{g^2}{4} \int d^4x A^\mu A^\nu A_\mu A_\nu) \rangle + \mathcal{O}(g^4)$$

s-channel:

$$(-ig^2/4)(\tilde{S}) \int d^4x A_1^\mu A^\mu A_2^\nu A^\nu A_3^\lambda A^\lambda A_4^\sigma A_\mu A_\nu$$

$$= (-ig^2/4)(\tilde{S}) \int d^4x \Delta(x_1 - x) \epsilon_1^\mu \epsilon_2^\nu \epsilon_3^\lambda \epsilon_4^\sigma \dots$$

$$= (-ig^2/4)(\tilde{S}) (\epsilon_1 \epsilon_3) (\epsilon_2 \epsilon_4)$$

$$\text{or } iV_{\mu\nu\rho\sigma} = ig^2 g_{\mu\rho} g_{\nu\sigma}$$

$\tilde{S}=4$  (Symmetry factor)

The 3pt Vertex

$$\langle A_1^\mu A_2^\nu A_3^\sigma \rangle \sim \left( -i\sqrt{2}g \int d^4x \left( \underbrace{\partial_\lambda A_1^\mu A_2^\nu A_3^\sigma}_{\text{3-gluon vertex}} \right) \right) \sim \frac{P_3^\mu \eta^{\nu\rho}}{P_3^2}$$

$$\text{For } A^\mu = \int \frac{d^4p}{(2\pi)^4} \epsilon^\mu(p) a^\dagger e^{ipx} + \dots$$

$$\partial^\lambda A^\rho = \int \frac{d^4p}{(2\pi)^4} \epsilon^\rho(p) (ik^\lambda) a^\dagger e^{ipx} + \dots$$

$$\Rightarrow \sim -i\sqrt{2}g (\epsilon_\mu \epsilon_\nu \epsilon_\sigma V^{\mu\nu\rho}(p_1, p_2, p_3))$$

For the 3-gluon vertex

$$V^{\mu_1 \mu_2 \mu_3}(p_1, p_2, p_3) = -\sqrt{2} (\eta^{\mu_1 \mu_2} p_3^{\mu_3} + \eta^{\mu_2 \mu_3} p_2^{\mu_1} + \eta^{\mu_3 \mu_1} p_1^{\mu_2})$$

$$\bullet A[123] = -\sqrt{2} ((\epsilon_1 \epsilon_2)(\epsilon_3 p_1) + (\epsilon_2 \epsilon_3)(\epsilon_1 p_2) + (\epsilon_3 \epsilon_1)(\epsilon_2 p_3))$$

Polarization Vectors:

$$\epsilon_-^\mu(p; q) = -\frac{\langle p | \gamma^\mu | q \rangle}{\sqrt{2} \langle p q \rangle}, \quad \epsilon_+^\mu = -\frac{\langle q | \gamma^\mu | p \rangle}{\sqrt{2} \langle q p \rangle}$$

• Massless systems  $P^\mu(P) = 0$

Let  $A_3[123] = A_3[1^- 2^- 3^+]$

$$*(E_1 E_2) = (E_1)_-^\mu (E_2)_{-\mu} = \left( \frac{-\langle 1 | \gamma^\mu | q_1 \rangle}{\sqrt{2} [1 q_1]} \right) \left( \frac{-\langle 2 | \gamma^\mu | q_2 \rangle}{\sqrt{2} [2 q_2]} \right)$$

$$\langle 1 |_\alpha \gamma^{\mu \alpha \dot{\alpha}} | q \rangle_\alpha \langle 2 |_\beta \gamma^{\mu \beta \dot{\beta}} | k \rangle_\beta$$

$$= 2 \langle 12 \rangle [qk]$$

$$(E_1)_-^\mu (E_2)_{-\mu} = \frac{\langle 12 \rangle [q_1 q_2]}{[1 q_1] [2 q_2]}$$

$$*(E_3)_+^\mu (P_3)_\mu = - \frac{\langle q_3 | \gamma^\mu | 3 \rangle}{\sqrt{2} \langle q_3 3 \rangle} (P_3)_\mu$$

Momentum conservation:  $\sum \langle q_i \rangle [ik] = 0$

$$= \langle q_3 | (-1) \rangle [11] [3] = \langle q_3 | \rangle [13]$$

$$\sim -\cancel{P} = |P\rangle [P] + |P\rangle \langle P|$$

(first term) =  $\left( \frac{\langle 12 \rangle [q_1 q_2]}{[1 q_1] [2 q_2]} \right) \left( \frac{\langle q_3 | \rangle [13]}{\sqrt{2} \langle q_3 3 \rangle} \right)$

Second term:  $(E_2)_-^\mu (E_3)_+^\mu (E_1)_-^\nu (P_2)_\nu$

$$(E_2)_-^\mu (E_3)_+^\mu = \left( \frac{-\langle 2 | \gamma^\mu | q_2 \rangle}{\sqrt{2} [2 q_2]} \right) \left( \frac{-\langle q_3 | \gamma^\mu | 3 \rangle}{\sqrt{2} \langle q_3 3 \rangle} \right)$$

$$= \frac{\langle 2 q_3 \rangle [q_2 3]}{[2 q_2] \langle q_3 3 \rangle}$$

$$(E_1)_-^\nu (P_2)_\nu = - \frac{\langle 1 | \gamma^\nu | q_1 \rangle (P_2)_\nu}{\sqrt{2} [1 q_1]} = \frac{\langle 12 \rangle [2 q_1]}{\sqrt{2} [1 q_1]}$$

(Second term) =  $\left( \frac{\langle 2 q_3 \rangle [q_2 3]}{[2 q_2] \langle q_3 3 \rangle} \right) \left( \frac{\langle 12 \rangle [2 q_1]}{\sqrt{2} [1 q_1]} \right)$

Third term:

$$(E_3)_+^\mu (E_1)_{-\mu} (E_2)_\nu (P_3)_\nu$$

By symmetry  $\sim \frac{\langle q_3 | \rangle [3 q_1] \langle 23 \rangle [3 q_2]}{\sqrt{2} [2 q_2] \langle q_3 3 \rangle [1 q_1]}$

$$A_3[1^- 2^- 3^+] = - \left( \langle 12 \rangle [q_1 q_2] \langle q_3 | \rangle [13] + \langle 2 q_3 \rangle [q_2 3] \langle 12 \rangle [2 q_1] + \langle q_3 | \rangle [3 q_1] \langle 23 \rangle [3 q_2] \right) \times \omega^{-1}$$

$$\omega \equiv [q_1 1] [q_2 2] \langle q_2 3 \rangle$$

Notes:  $\omega$  goes exactly as the helicity of each Gluon

$$h < 0 = \langle \cdot \cdot \rangle, \quad h > 0 = [\cdot \cdot]$$

\* For 3 particle kinematics we should have

$$p_1^\mu + p_2^\mu + p_3^\mu = 0$$

so at any vertex

$$\langle 12 \rangle [12] = 2 p_1^\mu p_{2\mu} = (p_1 + p_2)^2 = p_3^2 = 0$$

\* If  $|1\rangle \propto |2\rangle \propto |3\rangle$

Then  $A_3[1^- 2^- 3^+] = 0$ , so instead we choose kinematics s.t.  $|1\rangle \propto |2\rangle \propto |3\rangle$

$\Rightarrow$  First term vanishes  $[13] = 0$

$$A_3[1^- 2^- 3^+] = - \frac{\langle 2 q_2 \rangle [q_2 3] \langle 12 \rangle [2 q_1] + \langle q_2 1 \rangle [3 q_1] \langle 23 \rangle [3 q_2]}{[q_1 1] [q_2 2] \langle q_2 3 \rangle}$$

$$\begin{aligned} * \langle 12 \rangle [2 q_1] &= \langle 1 | 12 \rangle [2] [q_1] = - \langle 13 \rangle [3 q_1] \\ \Rightarrow &= \omega^{-1} [3 q_1] [q_2 3] (\langle 13 \rangle \langle 2 q_2 \rangle + \langle q_2 1 \rangle \langle 23 \rangle) \end{aligned}$$

\* Schouten identity:

$$\langle r | i \rangle \langle j k \rangle + \langle r | j \rangle \langle k i \rangle + \langle r | k \rangle \langle i j \rangle = 0$$

$$\langle 23 \rangle \langle q_2 1 \rangle + \langle 2 q_2 \rangle \langle 13 \rangle = - \langle 21 \rangle \langle 3 q_2 \rangle$$

$$A_3[1^- 2^- 3^+] = \omega^{-1} [3 q_1] [q_2 3] (- \langle 21 \rangle \langle 3 q_2 \rangle)$$

$$= - [3 q_1] [q_2 3] \langle 21 \rangle \langle 3 q_2 \rangle$$

$$\frac{[q_1 1] [q_2 2] \langle q_2 3 \rangle}{[q_1 1] [q_2 2]}$$

$$= [3 q_1] [q_2 3] \langle 21 \rangle$$

$$\frac{[q_1 1] [q_2 2]}{[q_1 1] [q_2 2]}$$

Via momentum conservation:

$$[q_1 3] \langle 23 \rangle = - [q_1 1] \langle 21 \rangle \quad \text{and}$$

$$[q_2 3] \langle 13 \rangle = - [q_1 1] \langle 12 \rangle$$

3-Gluon Amplitude:

$$A[1^- 2^- 3^+] = \frac{\langle 12 \rangle^2}{\langle 23 \rangle \langle 31 \rangle}$$

$$A_3[1^- 2^- 3^+] = \langle 23 \rangle \langle 31 \rangle$$

\* The opposite Helicity can be found via

$$A_3^*[1^- 2^- 3^+] = A_3[1^+ 2^+ 3^-]$$

Since

$$A^- \in \underline{SU(N)} \quad \text{and} \quad A^+ \in \underline{SUCM}$$

$$\langle 12 \rangle^* = [12] \quad [12]^2$$

$$\Rightarrow A_3[1^+ 2^+ 3^-] = [23][31]$$

Parke-Taylor n-gluon tree Amplitude: (PT)

$$A_n[1^+ \dots i^- \dots j^- \dots n^+] = \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$$

MHV Classification:

Gluon Amplitudes with two helicities and  $(n-2)$  hel. are MHV:  $2 \rightarrow (n-2)$  scattering



Applying Twistor Methods:  $A_n(1^+ 2^+, \dots, n^+) = 0$

Self-dual sector of YM is self consistent.

Classically integrable.

$n \geq 3$

$A_n[1^+ 2^+ \dots n^-] = 0$  (integrability of self dual  $\rightarrow$  No classical scattering)

\* Try to "derive" PT equation from twistor space  $PT$ .

(a) Take  $(n-2)$  pos. ( $h > 0$ ) helicity gluon reps  $f^{(+)}$

(b) Take 2 ( $h < 0$ ) gluons  $f^{(-)}$

(a):  $f^{(+)} \in H^{0,1}(PT, \mathcal{O}(0))$

(b):  $f^{(-)} \in H^{0,1}(PT, \mathcal{O}(-2))$

Put all of these on a single twistor line

$$X \cong \mathbb{CP}^1 \subset PT$$

and integrate over all positions of insertions and over (choice of  $X$ )  $\rightarrow$  over  $M^{\mathbb{C}}$

$$\int d^4x \int_{M^{\mathbb{C}} \times (\mathbb{CP})^n} \left( \prod_{k=1}^n \frac{D\lambda_k}{\langle \lambda_n \lambda_{k+1} \rangle} \right) \langle \lambda_i \lambda_j \rangle^4 \frac{f_i^{(-1)}}{f_j^{(-1)}} \Big|_x \prod_{l \neq ij} \frac{f_l^{(1)}}{f_l^{(1)}} \Big|_x \quad (178)$$

identify  $\lambda_{n+1} \sim \lambda_1$

Claim: The above Twistor space integral yields PT.

$\rightarrow$  Can evaluate using mon. eigenspace reps.

$$f^{(1)}|_x = \left( \frac{\langle \zeta \lambda \rangle}{\langle \zeta \kappa \rangle} \right)^{2m-1} \bar{\delta}(\langle \lambda \kappa \rangle) \exp \left[ \frac{i [\mu \kappa] \langle \zeta \kappa \rangle}{\langle \zeta \lambda \rangle} \right]$$

$$f_i^{(+1)}|_x = \left( \frac{\langle \zeta \lambda_i \rangle}{\langle \zeta \kappa_i \rangle} \right) \bar{\delta}(\langle \lambda_i \kappa_i \rangle) \exp \left[ \frac{i \langle \lambda_i | x | \kappa_i \rangle \langle \zeta \kappa_i \rangle}{\langle \zeta \lambda_i \rangle} \right]$$

$$f_j^{(-1)}|_x = \left( \frac{\langle \zeta \lambda \rangle}{\langle \zeta \kappa \rangle} \right) \bar{\delta}(\langle \lambda \kappa \rangle) \exp \left[ \frac{i \langle \lambda | x | \kappa \rangle \langle \zeta \kappa \rangle}{\langle \zeta \lambda \rangle} \right]$$

$$\int d^4x \left( \frac{D\lambda_1}{\langle \lambda_1 \lambda_2 \rangle} \dots \frac{D\lambda_n}{\langle \lambda_n \lambda_1 \rangle} \right) \langle \lambda_i \lambda_j \rangle^4 \left( \frac{\langle \zeta_i \lambda_i \rangle}{\langle \zeta_i \kappa_i \rangle} \right) \left( \frac{\langle \zeta_j \lambda_j \rangle}{\langle \zeta_j \kappa_j \rangle} \right) \bar{\delta}(\langle \lambda_i \kappa_i \rangle) \bar{\delta}(\langle \lambda_j \kappa_j \rangle) e^{\frac{i [\mu \kappa] \langle \zeta_i \kappa_i \rangle}{\langle \zeta_i \lambda_i \rangle}} \dots$$

$$\prod_{l \neq ij} \left( \frac{\langle \zeta_l \lambda_l \rangle}{\langle \zeta_l \kappa_l \rangle} \right) \bar{\delta}(\langle \lambda_l \kappa_l \rangle) e^{i \left( \frac{\langle \lambda_l | x | \kappa_l \rangle \langle \zeta_l \kappa_l \rangle}{\langle \zeta_l \lambda_l \rangle} \right)}$$

$$\Rightarrow \lambda_l^a \propto \kappa_l^a$$

$$\int d^4x \left( \frac{D\lambda_i}{\langle \lambda_i \lambda_{i+1} \rangle} \frac{D\lambda_j}{\langle \lambda_j \lambda_{j+1} \rangle} \langle \lambda_i \lambda_j \rangle^4 \right) \prod_{k \neq ij} D\lambda_k \bar{\delta}(\langle \lambda_k \kappa_k \rangle) (\dots)$$

$$\leadsto \langle \kappa | \gamma^\mu | \kappa \rangle = \kappa^\mu$$

$$\langle \kappa | P | \kappa \rangle = P^\mu \kappa_\mu$$

$$= \int d^4x \prod_{l \neq ij} \left( e^{i \frac{P^\mu \kappa_\mu}{\langle \zeta_l \lambda_l \rangle}} \right) = \prod_{l \neq ij} \delta(2P_l)$$

$$\hookrightarrow \left( \frac{1}{\langle 12 \rangle \dots \langle n1 \rangle} \right) \prod_{l \neq ij} \delta(P_l) \times \left( \frac{D\lambda_i}{\langle \lambda_i \lambda_{i+1} \rangle} \frac{D\lambda_j}{\langle \lambda_j \lambda_{j+1} \rangle} \frac{\delta(\langle \lambda_i \lambda_j \rangle)}{\langle \lambda_i \lambda_j \rangle^4} \right)$$

$$\begin{aligned}
 &= \left( \frac{1}{\langle 12 \rangle \dots \langle n1 \rangle} \right) \prod_{l \neq ij} \delta^4(p_l) \times \frac{1}{\langle i(i+1) \rangle \langle i(i+1) \rangle} \delta^4(p_i) \delta^4(p_j) \\
 &\quad \times \langle ij \rangle^4 \\
 &= \left( \frac{\langle ij \rangle^4}{\langle 12 \rangle \dots \langle n1 \rangle} \right) \prod_{l=1}^n \delta^4(p_l)
 \end{aligned}$$

Along a twistor  $X$  we have the MHV P.T. scattering equation when we expand in terms of

$$\int_{\mathcal{M}^c} \prod_{i=1}^n d\lambda_i \left| \langle \lambda_i \lambda_j \rangle^4 \prod_{l \neq ij} f_l^{(-1)} \right|_X \prod_{l \neq ij} f_l^{(+1)} \Big|_X = \frac{\prod_{l=1}^n \delta^4(p_l) \langle ij \rangle^4}{\langle 12 \rangle \dots \langle n1 \rangle}$$

(That's cool)

$f^{(h)} \in H^{0,1}(P\mathbb{T}, \mathcal{O}(2h-2))$   $\left| \begin{array}{l} X \cong \mathbb{CP}^1 \subset P\mathbb{T} \\ \text{(A line } X \text{ in } P\mathbb{T} \text{ corresponds to a } \mathcal{K} \text{ of } \mathcal{M}^c) \end{array} \right.$



\* Witten Conjecture: The entire tree-level S-matrix (4D) of YM could be written in terms of holomorphic maps from  $\mathbb{CP}^1 \rightarrow P\mathbb{T}$  (A-model String theory could obtain the ex)

$\hookrightarrow$  # of negative helicity gluons  $|g^-|$  is related to the degree of the map  $d$

$$d = |g^-| - 1$$

$\hookrightarrow$  MHV  $|g^-| = 2 \Rightarrow d = 1$  (linear map)

This is the incidence relation:  $v^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}$

$x^{\alpha\dot{\alpha}}$ : linearly (holomorphically) embedded Riemann sphere  $\mathbb{CP}^1 \hookrightarrow P\mathbb{T}$

$\hookrightarrow$  A degree- $d$  holomorphic map:  $Z^A(\sigma) \quad \sigma^a = (1, \sigma^1)$

$$Z^A(\sigma) = \mathcal{U}_{a_1 \dots a_d}^A \sigma^{a_1} \dots \sigma^{a_d}$$

$\sigma^{a_i}$  homogeneous on  $\mathbb{CP}^1$

$$Z^A: \mathbb{CP}^1 \hookrightarrow \mathbb{PT}$$

Claim:  $d=1$  the above is the incidence relations

$$Z^A(\sigma) = \mathcal{U}_{a_1}^A \sigma^{a_1}$$

$$\text{However } Z^A(r\sigma) = \mathcal{U}_{a_1}^A r\sigma^{a_1} = r Z^A(\sigma)$$

$$\text{So } Z^A \in \mathcal{O}(-1) \Rightarrow h = -\frac{1}{2}$$

$$Z^{\dot{a}}(\sigma) = \mathcal{U}_{a_1}^{\dot{a}} \sigma^{a_1} = \mathcal{U}^{\dot{a}a_1} \sigma_{a_1}$$

$\therefore \mathcal{U}^{\dot{a}a_1} \sim \chi^{\dot{a}a_1}$  for the case of  $d=1$

\* In general:

Rabin-Spradlin-Volkov formula: Made Witten conjecture

$$A_{n,d} = \int \frac{d^{4(d+1)} \mathcal{U}}{\text{Vol}(GL(2,\mathbb{C}))} \left( \prod_{\substack{k \in \mathbb{Z}^+ \\ k \leq d}} \left( \pi(\sigma_k, \bar{\sigma}_k) \right)^{u_k} \prod_{i=1}^n \sigma_i^{a_i} d\sigma_i \prod_{j \in \mathbb{Z}^+} f_j^{(c_j)}(Z(\sigma_j)) \right. \\ \left. (\sigma_i \sigma_{i+1}) \prod_{k \in \mathbb{Z}^+} f_k^{(t_k)}(Z(\sigma_k)) \right)$$

$\text{Vol}(GL(2,\mathbb{C}))$ : Means the moduli space is quotiented by a gauge redundancy.

$$* \text{Aut}(\mathbb{CP}^1) = \text{PSL}(2,\mathbb{C}) \cong GL(2,\mathbb{C})/\mathbb{C}^* *$$

\* Since  $A_{n,d}$  is an integral over the moduli space of maps  $\mathcal{U}: \mathbb{CP}^1 \hookrightarrow \mathbb{PT}$

$$M_{0,n}(\mathbb{PT}, d) \sim \frac{[A]}{GL(2,\mathbb{C})^*} \quad | \quad \tilde{A} \cong (\mathbb{CP}^1)^n$$

$\text{Vol}(GL(2,\mathbb{C}))$ : Removes the gauge redundancies. (i.e. introduce ghost fields).

$A_{n,d}$ : Computes every tree-level Scattering Amplitude in pure YM (4D).

General proof: look at how  $A_{n,d}$  degenerates when the underlying maps degenerate  
 $\rightarrow$  Boundary of  $M_{0,n}(\mathbb{PT}, d)$  ( $\partial M_{0,n}$ ) corresponds to nodal maps. These correspond to

# Factorization limits of $\mathcal{N}=4$

$\mathcal{N}=4$  SYM: on-shell:  $D=4$

$$S = \int d^4x \text{tr} \left( -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} (D\Phi_I)^2 + \frac{i}{2} \bar{\Psi} \not{D} \Psi + \frac{1}{2} \bar{\Psi} \Gamma^I [\Phi_I, \Psi] + \frac{g^2}{4} [\Phi_I, \Phi_J]^2 \right)$$

$$D_\mu = \partial_\mu - i [A_\mu, \cdot]$$

$\Phi_I$ : 6 real scalar fields, of global  $SO(6) \sim SU(4)$

$$I = 1, \dots, 6$$

R-symmetry

→ All fields transform in the Adjoint of  $SU(N)$

$$\text{i.e. } \frac{N^2-1}{N-1} \left\{ \begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} \right\}$$

\* 10D Majorana-Weyl:  $\Psi$ :

$\Gamma_I$ : 10D Clifford Algebra

→ Obtain by dimensional reduction of  $\mathcal{N}=1$  10D SYM

\* Group  $\Phi_I$  into six complex scalar fields  $\varphi^{AB} = -\varphi^{BA}$

$A, B = 1, 2, 3, 4$  of  $SU(4)$

$\varphi^{AB}$  is self dual  $\varphi_{AB} = \frac{1}{2} \epsilon_{ABCD} \varphi^{CD}$

→  $\varphi_{AB}$  is in  $\mathfrak{h}$  of  $SU(4)$  i.e.  $\underline{6}$

\* The 10D fermion fields give  $4+4$  gluino states

$$\left\{ \begin{array}{l} \lambda^A (h = +\frac{1}{2}) \\ \bar{\lambda}_A (h = -\frac{1}{2}) \end{array} \right\} \quad \square \quad \underline{4} \quad \text{transform in anti-fundamental}$$

\*  $\frac{g^2}{4} [\Phi_I, \Phi_J]^2$ : define the scalar Manifold ( $V_0 > 0$  brane

susy):  $[\Phi_I, \Phi_J] = 0$  Defines  $\mathcal{M}$ .

→ At the origin of  $\mathcal{M}$ :  $\langle \varphi^{AB} \rangle = 0$  (all other states vanish)

→ Massless theory / no running coupling i.e. it

flows freely to the IR and obtains a

Conformal symmetry

$\beta \sim 0$  to all orders

$$(SO(2,4) \cong SU(2,2))$$

→ Also invariant under Superconformal symmetry  $SU(2,2|4)$

$SU(2,2|4)$ : Bosonic

$$(SU(2,2), SU(4)_R)$$

$$\underline{15}$$

$$\underline{15}$$

$$20 \oplus 1$$

Fermionic:

$$(Q^{\alpha i}) \quad i = 1, \dots, 4$$

$$(SUSY)$$

$$32$$

\*  $[\bar{\Phi}_I, \Phi_J] = 0 \Rightarrow$  Theory is on the Coulomb branch (of the moduli space)

$\Rightarrow$  Work with  $N=4$  at the origin of the Moduli space.  
 look at  $d=4-\epsilon$ , then Conformal symmetry is broken  
 and the S-matrix is well defined.

Spectrum & susy Ward: (Massless!) (Quantized fields)

|            | $a$           | $a^A$                | $a^{AB}$           | $a^{ABC}$                                       | $a^{1234}$    |
|------------|---------------|----------------------|--------------------|-------------------------------------------------|---------------|
| " $\Phi$ " | 1 gluon $g^+$ | 4 gluons $\lambda^A$ | 6 scalars $S^{AA}$ | 4 gluons $\lambda^{ABC} \sim \tilde{\lambda}_0$ | 1 gluon $g^-$ |
| $h$        | 1             | $\frac{1}{2}$        | 0                  | $-\frac{1}{2}$                                  | -1            |

$A, B = 1, 2, 3, 4$  of  $SU(4)_R$

$\Rightarrow$  Transform under  $(2, -2h)$  of  $SU(4)$  (spin sums)

$\Rightarrow$  Theory is CPT self conjugate: The  $N=4$  susy  
 connects gluon states  $h = \pm 1$

\* Now need to write the states / Symmetry algebras  
 in terms of  $\{ \cdot \}$  &  $[ \cdot ]$   
 $\rightarrow$  Work out  $N=1$  first:

$$[a_{\pm}(p), a_{\pm}^{\dagger}(p')] = (2\pi)^3 E_p \delta^3(\vec{p} - \vec{p}')$$

$$\{b_{\pm}(p), b_{\pm}^{\dagger}(p')\} = (2\pi)^3 E_p \delta^3(\vec{p} - \vec{p}')$$

$$\left\{ \begin{array}{l} [P_L \psi_s(p) \rightarrow |p\rangle \quad s = + \\ \quad \quad \quad \rightarrow 0 \quad \quad s = - \\ [P_R \psi_s(p) \rightarrow |p\rangle \quad s = - \\ \quad \quad \quad \rightarrow 0 \quad \quad s = + \end{array} \right\} \quad \left| \begin{array}{l} P_L \psi_s \rightarrow |p\rangle, s = - \\ \vdots \end{array} \right.$$

Translate the field theory symmetries to Quantum  
 $N=1$ ):  $\delta_{\epsilon} \phi = \epsilon \psi$ ,  $\delta_{\epsilon} \bar{\phi} = \bar{\epsilon} \bar{\psi}$   $\phi \in \mathbb{C}$

$$\delta_{\epsilon} \psi_a = -i \sigma_{\alpha\dot{\beta}}^{\mu} \bar{\epsilon}^{\dot{\beta}} \partial_{\mu} \phi$$

$$\delta_{\epsilon} \bar{\psi}_{\dot{a}} = i \partial_{\mu} \bar{\phi} \epsilon^{\dot{b}} \sigma_{\dot{b}a}^{\mu}$$

The QFT has:

$$\phi(x) = \int d^3p [a_{-}(p) e^{ipx} + a_{+}^{\dagger}(p) e^{-ipx}]$$

$$\psi_{\alpha}(x) = \sum_{s=\pm} \int d^3p [b_s(p) P_L \psi_s(p) e^{ipx} + b_s^{\dagger} P_L \psi_s(p) e^{-ipx}]$$

$$\psi^\dagger(x) \equiv \bar{\psi} = \bar{\psi} \gamma^0 = \bar{\psi} \gamma^0$$

$$\delta_e \phi = e \phi$$

$$\begin{aligned} \rightarrow \int d^3p (\delta_e a_-(p) e^{ipx} + \delta_e a_+^\dagger(p) e^{-ipx}) \\ = \sum_{s=\pm} \int d^3p (b_s(p) e^\alpha (P_L \chi_s)_\alpha e^{ipx} + b_s^\dagger P_L \chi_s e^{-ipx}) \end{aligned}$$

matching

$$\begin{aligned} \delta_e a_-(p) &= [E p] b_-(p) \\ \delta_e a_+(p) &= \langle E p \rangle b_+(p) \end{aligned} \quad \left| \begin{aligned} \langle E \rangle_i &= E_i^\dagger \\ [E]_\alpha &= E_\alpha \end{aligned} \right.$$

$$\text{For } \delta_e \phi_\alpha = -i \sigma_{\alpha\dot{\beta}}^\mu E^\dagger_{\dot{\beta}} \partial_\mu \phi$$

$$\text{LHS: } \sum_{s=\pm} \int d^3p \delta_e b_s(p) (P_L \chi_s)_\alpha e^{ipx} + \dots$$

$$\Rightarrow \delta_e b_-(p)$$

$$\text{RHS: } \int d^3p a_-(p) (-i \sigma_{\alpha\dot{\beta}}^\mu E^\dagger_{\dot{\beta}} (i p_\mu)) e^{ipx} + \dots$$

$$\Rightarrow \delta_e b_-(p) (P_L \chi_-)_\alpha = a_-(p) \sigma_{\alpha\dot{\beta}}^\mu E^\dagger_{\dot{\beta}} p_\mu$$

$$\delta_e b_-(p) (P_L \chi_-)_\alpha = a_-(p) \langle E \rangle p_\alpha$$

$$p = -|p\rangle [p] \langle p|$$

$$\delta_e b_-(p) = \langle E p \rangle a_-(p)$$

$$\delta_e b_+(p) = [E p] a_+(p)$$

Field  $\xrightarrow{\text{SUSY}}$  QFT

\*Can show closure:  $\delta_{e_1} \delta_{e_2} a_-(p)$

$$\delta_{e_1} ([E_2 p] b_-(p)) = [E_2 p] \langle E_1 p \rangle a_-(p)$$

$$\delta_{e_2} \delta_{e_1} a_-(p) = [E_1 p] \langle E_2 p \rangle a_-(p)$$

$$\Rightarrow [\delta_1, \delta_2] a_-(p) = ([E_2 p] \langle E_1 p \rangle - [E_1 p] \langle E_2 p \rangle) a_-(p)$$

$$= -([E_2 p] \langle p E_1 \rangle - \langle E_2 p \rangle [p E_1]) a_-(p)$$

$$= ([E_2 (-|p\rangle \langle p|)] |E_1\rangle - \langle E_2 (-|p\rangle \langle p|) |E_1\rangle) a_-(p)$$

$$= (E^\alpha p_{\alpha\dot{\beta}} E^\dagger_{\dot{\beta}} - E^\dagger_{\dot{\beta}} p^{\alpha\dot{\beta}} E_\alpha) a_-(p)$$

$$= (E^\alpha \gamma_{\alpha\dot{\beta}}^\mu E^\dagger_{\dot{\beta}} - E^\dagger_{\dot{\beta}} (\gamma^{\alpha\dot{\beta}})^\mu E_\alpha) p_\mu a_-(p)$$

$$a^\mu = E^\alpha \gamma_{\alpha\dot{\beta}}^\mu E^\dagger_{\dot{\beta}} - E^\dagger_{\dot{\beta}} (\gamma^{\alpha\dot{\beta}})^\mu E_\alpha$$

$$\therefore [\delta_1, \delta_2] a_-(p) = a^\mu p_\mu a_-(p)$$

Can write  $a^\mu$  in terms of Majorana

$$a^\mu = E_{\mu,2} \gamma^\mu E_{\mu,1}$$

$$E_\mu = \begin{pmatrix} E_\alpha \\ E^\dagger_{\dot{\alpha}} \end{pmatrix}$$

$$= \begin{pmatrix} [E]_\alpha \\ \langle E \rangle_{\dot{\alpha}} \end{pmatrix}$$

Can also package up the sys generators  $([E], \dots)$

$$Q_M \equiv \begin{pmatrix} [Q]_\alpha \\ [Q^\dagger]^\alpha \end{pmatrix} \quad \bar{E}_M \sim ([E]^\alpha, [E]_\alpha)$$

Can deduce their bracket form from

$$\delta_\epsilon \psi = [\bar{E}_M Q_M, \psi] = [([E]_\alpha + \langle E \rangle_\alpha), \psi]$$

\* Start with  $\delta_\epsilon a_-(p) = [E]_\alpha b_-(p)$

$$\begin{aligned} \delta_\epsilon a_-(p) &= [\langle E \rangle_\alpha + [E]_\alpha, a_-(p)] \\ &= E_\alpha [Q^\dagger_\alpha, a_-] + E^\alpha [Q_\alpha, a_-(p)] \end{aligned}$$

Since  $\sim [E]_\alpha \Rightarrow [Q^\dagger_\alpha, a_-(p)] = 0$

$$\delta_\epsilon a_-(p) = E^\alpha [Q_\alpha, a_-(p)] =$$

$$Q_\alpha \propto [P]_\alpha b_-(p)$$

However, assuming

$$Q_\alpha \sim a [\tilde{P}]_\alpha b_-(p)$$

$$\delta_\epsilon a_-(p) = a [E \tilde{P}] [b_-(p), a_-(p)] = 0$$

trivially, so  $a$  above must be  $a^\dagger_-(p)$

i.e.  $Q_\alpha \sim [\tilde{P}]_\alpha a_-(\tilde{p}) b_-(p)$

$$\begin{aligned} \delta_\epsilon a_-(p) &= [E \tilde{P}] b_-(\tilde{p}) [a^\dagger_-(\tilde{p}), a_-(p)] \\ &= -[E \tilde{P}] b_-(\tilde{p}) (2\pi)^3 2E_p \delta^3(\vec{p} - \vec{\tilde{p}}) \end{aligned}$$

finally, we see that  $Q_\alpha$  must be expanded over the measure  $\int \tilde{d}\vec{p}$  i.e. a full Quantum field

$$Q_\alpha \sim - \int \tilde{d}\vec{p} [\tilde{P}]_\alpha b_-(\tilde{p}) a^\dagger_-(\tilde{p})$$

\* Doing the same analysis for

$$\delta_\epsilon b_+(p) \text{ we determine } Q_\alpha \sim \int \tilde{d}\vec{p} [\tilde{P}]_\alpha (a_+(p) b^\dagger_+(p))$$

and so we conclude.

$$\begin{aligned} [Q]_\alpha &= \int \tilde{d}\vec{p} [P]_\alpha (a_+(p) b^\dagger_+(p) - b_-(p) a^\dagger_-(p)) \\ [Q^\dagger]^\alpha &= \int \tilde{d}\vec{p} [P]^\alpha (a_-(p) b^\dagger_-(p) - b_+(p) a^\dagger_+(p)) \end{aligned}$$

→ find  $[Q^\dagger]^\alpha$  from analysis of  $\delta_\epsilon a \neq \delta_\epsilon \psi$ .

$$\{Q_\alpha, Q^\dagger_\beta\} = -2\sigma_{\alpha\beta} P_M$$

→ Can show Quantized version:

$$\{[Q]_\alpha, \langle Q^\dagger |_{\tilde{\beta}}\} \sim P_{\alpha\tilde{\beta}}$$

$$\int d^3p \int d^3k [P]_\alpha \langle k |_{\tilde{\beta}} \{ a_+(p) b_+^\dagger(p) - b_-(p) a_-^\dagger(p), a_-(k) b_-^\dagger(k) - \}$$

$$(c) \quad a_+(p) b_+^\dagger(p) a_-(k) b_-^\dagger(k) + a_-(k) b_-^\dagger(k) a_+(p) b_+^\dagger(p) = 0$$

(all commute)

$$(ic) \quad \{ a_+(p) b_+^\dagger(p), (-1) b_+(k) a_+^\dagger(k) \}$$

$$= - (a_+(p) b_+^\dagger(p) b_+(k) a_+^\dagger(k) + b_+(k) a_+^\dagger(k) a_+(p) b_+^\dagger(p))$$

$$= - [ (a_+^\dagger(k) a_+(p) + (2\pi)^3 2E_p \delta^3(p-k)) b_+^\dagger(p) b_+(k) + \hat{1} ]$$

$$= - [ a_+^\dagger(k) a_+(p) (-b_+(k) b_+^\dagger(p) + (2\pi)^3 2E_p \delta^3(p-k))$$

$$+ (2\pi)^3 2E_p \delta^3(p-k) (-b_+(k) b_+^\dagger(p) + (2\pi)^3 2E_p \delta^3(p-k))$$

$$+ \dots ]$$

$$= - [ -a_+^\dagger(k) b_+(k) a_+(p) b_+^\dagger(p) + a_+^\dagger(k) a_+(p) (2\pi)^3 2E_p \delta^3$$

$$- (2\pi)^3 2E_p \delta^3 b_+(k) b_+^\dagger(p) + ((2\pi)^3 2E_p \delta^3)^2 + \dots ]$$

$$= - [ a_+^\dagger(k) a_+(p) \delta^3 - b_+(k) b_+^\dagger(p) \delta^3 + (\delta^3)^2 ]$$

$\leadsto$  flip  $b$

$$= - [ (a_+^\dagger(k) a_+(p) + b_+^\dagger b_+) \delta^3 + (\delta^3)^2 ]$$

$$(ic) \quad \{ b_-(p) a_-^\dagger(p), b_+(k) a_+^\dagger(k) \} = 0$$

$$(cn) = - \{ b_-(p) a_-^\dagger(p), a_-(k) b_-^\dagger(k) \}$$

Same analysis as (ci)

$$= - (a_-^\dagger(k) a_-(p) \delta^3 + b_-^\dagger(p) b_-(k) \delta^3 - (\delta^3)^2)$$

$$\{[Q]_\alpha, \langle Q^\dagger |_{\tilde{\alpha}}\} = P_{\alpha\tilde{\alpha}} \sum_{i,\pm} N_{i\pm}$$

where

$$N_{i\pm} = \int d^3p a_{i\pm}^\dagger(p) a_{i\pm}(p) \quad (\text{Number operator})$$

On the annihilation operators:

$$[Q_\alpha, a_-(p)] = [P]_\alpha b_-(p) \quad [Q^{\dagger\tilde{\alpha}}, b_-(p)] = |p\rangle^{\tilde{\alpha}} a_-(p)$$

$$[Q_\alpha, b_-(p)] = 0 \quad [Q^{\dagger\tilde{\alpha}}, a_-(p)] = 0$$

$$[Q_\alpha, b_+(p)] = [P]_\alpha a_+(p) \quad [Q^{\dagger\tilde{\alpha}}, a_+(p)] = |p\rangle^{\tilde{\alpha}} b_+(p)$$

$$[Q_\alpha, a_+(p)] = 0 \quad [Q^{\dagger\tilde{\alpha}}, b_+(p)] = 0$$

$\hookrightarrow$  1-particle states exist in a (super) multiplet

$$\phi: \quad h=0, \quad b_-: h=-1/2, \quad b_+: h=+1/2$$

$Q_\alpha$ : lowers helicity by  $-1/2$

$Q^{\dagger 0}$ : raises " "  $+1/2$

$\Rightarrow$  Since  $[P^\mu, Q_a] = [P^\mu, Q^{\dagger a}] = 0$ , states in the same multiplet have the same mass.

\*  $N \Rightarrow$  R-symmetry of  $SU(N)$

$N=2$  SYM:  $(\pm \frac{1}{2}, \pm 1)$ ,  $2^N$  states

$\hookrightarrow$  Take  $N=2$  chiral Model with interactions

$$\mathcal{L}_I = \frac{i}{2} g \phi \psi \psi + \frac{i}{2} g^* \bar{\phi} \psi^\dagger \psi^\dagger - \frac{1}{4} |g|^2 |\phi|^4$$

The 4pt tree Amplitudes are given by

$$A_u^{\text{tree}}(\phi \phi \bar{\phi} \bar{\phi}) = -|g|^2, A_u^{\text{tree}}(\phi f \bar{f} \bar{\phi}) = -|g|^2 \frac{\langle 24 \rangle}{\langle 34 \rangle}$$

$$A_u^{\text{tree}}(f^- f^- f^+ f^+) = |g|^2 \langle 12 \rangle / \langle 34 \rangle$$

•  $A_u^{\text{tree}}(f^- f^- f^+ f^+)$ : This is

$$\begin{aligned} & \langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle : \text{Contract at second order in} \\ & \langle \psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4 \rangle \sim \left\langle \frac{i}{4} |g|^2 \int d^4 x d^4 z (\bar{\psi}_1 \psi_2 \bar{\psi}_3 \psi_4) \phi \psi \psi \phi^\dagger \psi^\dagger \psi^\dagger \right\rangle \\ & \sim \left( \frac{i}{4} g^2 \right) \bar{u}_4 v_3 \left( \frac{-i}{(p_1 + p_2)^2} \right) \bar{u}_2 v_1 \end{aligned}$$

(s-channel)

•  $\bar{u}_4 v_3$ : must have the same helicity  $h = +1/2$

$\bar{u}_2 v_1$ : " "  $h = -1/2$

$$\sim \frac{i}{4} |g|^2 [43] \frac{(-i)}{(p_1 + p_2)^2} \langle 21 \rangle = \frac{i}{4} |g|^2 \langle 12 \rangle \frac{[34]}{\langle 12 \rangle [12]}$$

• Sum over other channels:

$$A_u^{\text{tree}}(f^- f^- f^+ f^+) = |g|^2 \frac{[34]}{[12]}$$

By momentum conservation

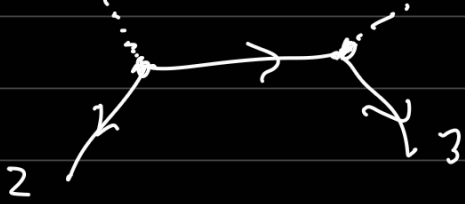
$$\langle 12 \rangle [12] = (p_1 + p_2)^2 = (p_3 + p_4)^2 = \langle 34 \rangle [34]$$

$$\text{or } \frac{[34]}{[12]} = \frac{\langle 12 \rangle}{\langle 34 \rangle}$$

$$\therefore A_u^{\text{tree}}(f^- f^- f^+ f^+) = |g|^2 \langle 34 \rangle \quad (\text{as claimed})$$

•  $A_u^{\text{tree}}(\phi \bar{\phi} f^+ f^-)$ :

$$\langle \phi \bar{\psi} \psi \bar{\phi} \rangle \sim |g|^2 \int d^4 x d^4 z \langle \phi \bar{\psi} \psi \bar{\phi} (\phi \psi \psi) (\bar{\phi} \psi^\dagger \psi^\dagger) \rangle$$



$$\sim i g^2 \bar{u}(p_2) \left( \frac{(p_1 + p_2)_\mu \gamma^\mu}{(p_1 + p_2)^2} \right) u(p_3)$$

$$\begin{aligned} & \sim g^2 \langle 2 | (p_1 + p_2) | 3 \rangle / \langle 12 \rangle [12] \\ & = g^2 \langle 2 | p_2 | 3 \rangle / \langle 12 \rangle [12] \\ & = + g^2 \frac{\langle 12 \rangle [13]}{\langle 12 \rangle [12]} = + g^2 \frac{[13]}{[12]} \end{aligned}$$

\* Claim:  $\langle 34 \rangle [13] = -\langle 24 \rangle [12]$

proof:  $\langle 4 | ( ) | 1 \rangle = 0$

$$\langle 42 \rangle [21] + \langle 43 \rangle [31] = 0 \quad (\text{closed})$$

$$\langle 24 \rangle [12] + \langle 34 \rangle [13] = 0 \Rightarrow \langle 34 \rangle [13] = -\langle 24 \rangle [12]$$

$$\text{so } [13] / [12] = -\langle 24 \rangle / \langle 34 \rangle$$

$$\therefore A_u^{\text{tree}}(\phi f^- f^+ \bar{\phi}) = -g^2 \frac{\langle 24 \rangle}{\langle 34 \rangle}$$

\* Note:  $A_u^{\text{tree}}(\phi, f^- f^+ \bar{\phi}) = \frac{\langle 24 \rangle}{\langle 34 \rangle} A_u^{\text{tree}}(\phi \phi \bar{\phi} \bar{\phi})$

$$A_u^{\text{tree}}(f^- f^- f^+ f^+) = -\frac{\langle 12 \rangle}{\langle 24 \rangle} A_u^{\text{tree}}(\phi f^- f^+ \bar{\phi})$$

These hold, by supersymmetry, not only at tree level but to all orders of pert. theory.

Supersymmetric Ward Identities:

For all outgoing particles:  $\langle 0 | \mathcal{Q}_1(p_1) \dots \mathcal{Q}_n(p_n) | 0 \rangle$

where  $\mathcal{Q}_i(p_i)$  are annihilation ops that act on the out

ex:  $A_u(\phi f^- f^+ \bar{\phi}) = \langle 0 | u_-(p_1) b_-(p_2) b_+(p_3) a_+(p_4) | 0 \rangle$

Suppose the vacuum is susy invariant:

$$Q|0\rangle = 0 = Q^\dagger|0\rangle \quad (\text{Clifford vacuum})$$

then

$$\langle 0 | [Q^\dagger, \mathcal{Q}_1(p_1) \dots \mathcal{Q}_n(p_n)] | 0 \rangle = 0$$

$$= \langle 0 | Q^\dagger ( ) - ( ) Q^\dagger | 0 \rangle$$

$$= \langle 0 | (Q^\dagger \mathcal{Q}_1(p_1)) \mathcal{Q}_2 \dots - ( \mathcal{Q}_1 \dots ) (\mathcal{Q}_n Q^\dagger) | 0 \rangle$$

$$= \langle 0 | (\theta_1 Q^\dagger \pm [\theta_1^\dagger, \theta_1]) \theta_2 \dots - ( ) Q^\dagger | 0 \rangle$$

(Supersymmetry Word Identity)

$$= \sum_{i=1}^n (-1)^{\sum_{j=1}^{i-1} |\theta_j|} \langle 0 | \theta_1(p_1) \dots [\theta_i^\dagger, \theta_i(p_i)] \dots \theta_n(p_n) | 0 \rangle$$

$$\therefore \langle 0 | [\theta_1^\dagger, \theta_1 \dots \theta_n] | 0 \rangle = 0$$

$|\theta_i| = 0$  (Bosonic) (Valid at any

$|\theta_i| = 1$  (Fermionic)

Take  $A_u(\phi f^- \bar{\phi})$

$$\begin{aligned} 0 &= \langle 0 | [\theta_1^\dagger, a_-(p_1) b_-(p_2) a_+(p_3) a_+(p_4)] | 0 \rangle \\ &= \langle 0 | [\theta_1^\dagger, a_-(p_1)] b_-(p_2) a_+(p_3) a_+(p_4) | 0 \rangle \\ &\quad + \langle 0 | a_-(p_1) [\theta_1^\dagger, b_-(p_2)] a_+(p_3) a_+(p_4) | 0 \rangle \\ &\quad + \langle 0 | a_-(p_1) b_-(p_2) [\theta_1^\dagger, a_+(p_3)] a_+(p_4) | 0 \rangle \\ &\quad - \langle 0 | a_-(p_1) b_-(p_2) a_+(p_3) [\theta_1^\dagger, a_+(p_4)] | 0 \rangle \\ &= |2\rangle \langle 0 | a_-(p_1) a_-(p_2) a_+(p_3) a_+(p_4) | 0 \rangle \\ &\quad - |3\rangle \langle 0 | a_-(p_1) b_-(p_2) b_+(p_3) a_+(p_4) | 0 \rangle \\ &\quad - |4\rangle \langle 0 | a_-(p_1) b_-(p_2) a_+(p_3) b_+(p_4) | 0 \rangle \end{aligned}$$

or

$$0 = |2\rangle A_u(\phi \bar{\phi} \bar{\phi}) - |3\rangle A_u(\phi f^- f^+ \bar{\phi}) - |4\rangle A_u(\phi f^- \bar{\phi} f^+)$$

SWI encodes two relationships since  $Q^\dagger \rightarrow Q$

→ Projecting out the  $|4\rangle$  state

$$\left( \frac{\langle 2u |}{\langle 23u |} \right) A_u(\phi \bar{\phi} \bar{\phi}) = A_u(\phi f^- f^+ \bar{\phi})$$

→ This holds at any loop order.

Projecting out  $|2\rangle$ :

$$A_u(\phi f^- \bar{\phi} f^+) = - \frac{\langle 23 |}{\langle 2u |} A_u(\phi f^- f^+ \bar{\phi})$$

(With these no further information can be gained)

\* Testing the Word Identity:

$$i) \langle 0 | b_-(p_1) a_-(p_2) b_+(p_3) b_-(p_4) | 0 \rangle : A_u(f^- \phi f^+ f^-)$$

$$0 = |1\rangle A_u(\phi \phi f^+ f^-) - |4\rangle A_u(f^- \phi f^+ f^-)$$

$$ii) \langle 0 | b_-(p_1) b_-(p_2) b_+(p_3) a_+(p_4) | 0 \rangle : A_u(f^- f^- f^+ \bar{\phi})$$

$$\begin{aligned} 0 &= |1\rangle A(\phi f^- f^+ \bar{\phi}) - |2\rangle A(f^- \phi f^+ \bar{\phi}) \\ &\quad - |4\rangle A_u(f^- f^- f^+ f^+) = 0 \end{aligned}$$

projecting out the  $|2\rangle$  state:

$$A_n(f^- f^- f^+ f^+) = -\frac{\langle 12 \rangle}{\langle 24 \rangle} A_n(4 f^- f^+ \bar{\phi}) \quad (\text{as above})$$

Higher order scattering: take 6 pt

$$0 = \langle 0 | [Q^\dagger, a_1 - a_2 - b_3 - a_4 + a_5 + a_6] | 0 \rangle$$

$$\hookrightarrow A_6(\phi\phi f^- \bar{\phi}\bar{\phi}\bar{\phi})$$

$$0 = \langle 3 \rangle A_6(\phi\phi\phi\bar{\phi}\bar{\phi}\bar{\phi}) - \langle 4 \rangle A_6(\phi\phi f^- f^+ \bar{\phi}\bar{\phi}) - \langle 5 \rangle A_6(\phi\phi f^- \bar{\phi} f^+ \bar{\phi}) - \langle 6 \rangle A_6(\phi\phi f^- \bar{\phi}\bar{\phi} f^+)$$

: Have two relationships but four unknowns  $A$

\* Now extend the theory to  $N=1$  SYM

$$\mathcal{L} = -\frac{1}{4} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + i \text{Tr}(\lambda^\dagger \sigma^\mu D_\mu \lambda)$$

• All fields in  $SU(N)$  adjoint.

$\tilde{Q}$ : raises helicity by  $+1/2$

$Q$ : lowers " "  $-1/2$

$$\tilde{Q}|0\rangle = +1/2$$

$$|4\rangle$$

$$\rightarrow Q|4\rangle$$

$$-1/2$$

$C_\pm$ : gluon ann. operators  $h = \pm 1$

$b_\pm$  gluinos ann " "  $h = \pm 1/2$

$$\Rightarrow \begin{array}{l|l} [\tilde{Q}, C_\pm(i)] = 0 & [Q, C_\pm(i)] = [i| b_\pm(i) \\ [\tilde{Q}, C_\pm(i)] = [i| b_\mp(i) & [Q, C_\mp(i)] = 0 \\ [\tilde{Q}, b_\pm(i)] = [i| C_\mp(i) & [Q, b_\pm(i)] = 0 \\ [\tilde{Q}, b_\mp(i)] = 0 & [Q, b_\mp(i)] = [i| C_\mp(i) \end{array}$$

\* The SWI holds here as well:

$$\langle 0 | [\tilde{Q}, b_1 + C_2 + \dots + C_n] | 0 \rangle = 0$$

$$= |1\rangle A_n[g^- g^+ \dots g^+] = 0$$

$\Rightarrow$  All given amplitude with +- variables to all orders in pert. theory.

$$\langle 0 | [\tilde{Q}, b_- + C_2 + \dots + C_n] | 0 \rangle = 0$$

$$0 = [1| A_n(g^- g^+ \dots g^+) = 0$$

$$\text{SYM: } A_n^{\text{L-loop}}(g^+ \dots g^+) = A_n^{\text{L-loop}}(g^- g^+ \dots g^+) = 0$$

$\hookrightarrow$  Holds true at tree level in YM.

$A=3$ : Differs mod  $A_3(g^- g^+ g^+) \neq 0$  3-particle special

$\hookrightarrow$  Kinematics

$$\text{* In SYM } A_n(g^- g^- \lambda^+ g^+ \dots g^+) = \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n(g^- g^+ g^+ g^+ \dots g^+)$$

"proof": Use SYM Ward:

$$A_n [g^- g^- \lambda^+ g^+ \dots g^+]$$

$$[\tilde{Q}, c_+] = 0, \quad [\tilde{Q}, b_+] = |i\rangle c_+$$

$$[\tilde{Q}, c_-] = |i\rangle b_-$$

$$\Rightarrow 0 = |1\rangle A_n [\lambda^- g^- \lambda^+ g^+ \dots g^+] + |2\rangle A_n [g^- \lambda^- \lambda^+ g^+ \dots g^+] - |3\rangle A_n [g^- g^- g^+ g^+ \dots g^+]$$

Projecting out  $|1\rangle$ :

$$A_n [g^- \lambda^- \lambda^+ g^+ \dots g^+] = \frac{\langle 12 \rangle}{2 \langle 12 \rangle} A_n [g^- g^- g^+ g^+ \dots g^+]$$

\* Next we try to relate  $A_n [\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+]$  to the MHV amplitude:

$A_n [g^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+]$  into Ward

$$0 = |1\rangle A_n [\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+] - |3\rangle A_n [g^- \lambda^- g^+ \lambda^+ g^+ \dots g^+] + |4\rangle A_n [g^- \lambda^- \lambda^+ g^+ \dots g^+]$$

Project out  $|3\rangle$

$$A_n [\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+] = \left( \frac{\langle 13 \rangle}{\langle 34 \rangle} \right) \left( \frac{\langle 12 \rangle}{\langle 12 \rangle} \right) A_n [g^- g^- g^+ \dots g^+]$$

$$A_n [\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+] = \frac{\langle 13 \rangle^2}{\langle 34 \rangle \langle 12 \rangle} A_n [g^- g^- g^+ \dots g^+]$$

This makes sense, since I can take

$$A_n \sim A_{n0}$$

\* In certain Susy models we can build higher pt amplitudes from lower pts. (Easily seen in super space)

Back to  $N=4$  ( $A, B = 1, \dots, 4$  of  $su(4)_R$ )

$$[\tilde{Q}_A, a(i)] = 0$$

$$[\tilde{Q}_A, a^B(i)] = |i\rangle \delta_A^B a(i)$$

$$[\tilde{Q}_A, a^{BC}(i)] = |i\rangle 2! \delta_A^{[B} a^{C]}(i)$$

$$[\tilde{Q}_A, a^{BCD}(i)] = |i\rangle 3! \delta_A^{[B} a^{CD]}(i)$$

$$[\tilde{Q}_A, a^{BCDE}(i)] = |i\rangle 4! \delta_A^{[B} a^{CDE]}(i)$$

$$[Q^A, a(i)] = [i| a^A(i)$$

$$[Q^A, a^B(i)] = [i| a^{AB}(i)$$

$$[Q^A, a^{BC}(i)] = [i| a^{ABC}(i)$$

$$[Q^A, a^{BCD}(i)] = [i| a^{ABCD}(i)$$

$$[Q^A, a^{1234}(i)] = 0$$

$\tilde{Q}_A = Q_A^\dagger$ : Raises (+1/2 h) and removes an index

\* From SWI: Start from  $A_n [g^- g^- \lambda^4 g^+ \dots g^+]$

in  $N=4$

$$Q^{\tilde{A}} = Q^{\tilde{A}} + \dots$$

$$g : a : \text{Gluon} (-1)$$

$$(1)^4 : \text{Gluon} (+1/2)$$

$$0 = |1\rangle (4!) \delta_A^{11} \langle 0 | a^{234} \dots | 0 \rangle \\ + |2\rangle (4!) \delta_A^{11} \langle 0 | a^{234} \dots | 0 \rangle \\ + |3\rangle \delta_A^{11} \langle 0 | g^- g^- a \dots | 0 \rangle$$

$\uparrow$   $g^+$

Note: the final term vanishes for  $A \neq 4$   
and so to obtain the full term above  $A=4$

$$\therefore 4! \delta_A^{11} a^{234} = -a^{123}$$

$$0 = -|1\rangle A_n [\lambda^{123} g^- \lambda^4 g^+ \dots g^+] - |2\rangle A_n [g^- \lambda^{123} \lambda^4 g^+ \dots g^+] \\ + |3\rangle A_n [g^- g^- g^+ g^+ \dots g^+]$$

or, projecting out  $|1\rangle$ :

$$* A_n [g^- \lambda^{123} \lambda^4 g^+ \dots g^+] = \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n [g^- g^- g^+ g^+ \dots g^+]$$

\* Next, we want to look at scalar gluon scattering

→ Start with  $A_n [g^- \lambda^{123} s^{34} g^+ \dots g^+]$

$$0 = \langle 0 | [\tilde{Q}, \dots] | 0 \rangle \\ = -|1\rangle A_n [\lambda^{123} (s^{34} g^+ \dots g^+)] \\ + |2\rangle A_n [g^- s^{12} s^{34} g^+ \dots g^+] \\ - |3\rangle 2! \delta_A^{11} \langle 0 | a^{234} \dots \rangle$$

$$[\tilde{Q}_A, a^{123}] = |1\rangle 3! \delta_A^{11} a^{234} \sim s^{12}$$

so  $A=3$

$$0 = |1\rangle A_n (\lambda^{124} \dots) - |2\rangle A_n [g^- s^{12} s^{34} g^+ \dots g^+] \\ + |3\rangle A_n [g^- \lambda^{123} \lambda^4 g^+ \dots g^+]$$

Project out  $|1\rangle$

$$A_n [g^- s^{12} s^{34} g^+ \dots g^+] = \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n [g^- \lambda^{123} \lambda^4 g^+ \dots g^+] \\ = \frac{\langle 13 \rangle^2}{\langle 12 \rangle^2} A_n [g^- g^- g^+ g^+ \dots g^+]$$

\*  $N=4$  SYM:

$$* A_n [g^+ g^+ \dots g^- \dots g^- \dots g^+] = \frac{\langle i j \rangle^4}{\langle 12 \rangle^4} A_n [g^- g^- g^+ g^+ \dots g^+] *$$

• Super Ward cannot mix gluon amplitudes with different  $K$  in the  $N^k \text{MHV}$  classification.

→ Given tree amplitudes don't "see" the susy partners  
N=4 MHV Classification:

→ Given scattering with  $k+2$  negative helicity states  $E \cong \mathbb{S}^4$

On-shell Superspace:  $\mathcal{H}_A: \text{SU}(4)$ ,  $A=1,2,3,4$

$$\Omega \equiv g^+ + \eta_A \lambda^A - \frac{1}{2} \eta_A \eta_B \sigma^{AB} - \frac{1}{3!} \eta_A \eta_B \eta_C \lambda^{ABC} + \eta_1 \eta_2 \eta_3 \eta_4 g^-$$

$E$   
 $\downarrow$   
 $\text{PT}$

→  $\Omega \in \mathcal{H}^{0,1}(\text{PT}, \mathcal{O}(0) \otimes \text{End}(E))$

$N=4$ :

|          |       |                |                             |                                          |                                                       |
|----------|-------|----------------|-----------------------------|------------------------------------------|-------------------------------------------------------|
| particle | $g^+$ | $\lambda^A$    | $\sigma^{AB}$               | $\lambda^{ABC}$                          | $g^- \equiv g^{1234}$                                 |
| operator | $1$   | $\partial_i^A$ | $\partial_i^A \partial_i^B$ | $\partial_i^A \partial_i^B \partial_i^C$ | $\partial_i^1 \partial_i^2 \partial_i^3 \partial_i^4$ |

\* This has on-shell supercharges:

$q^{Aa} \equiv [P]^a \frac{\partial}{\partial \eta_A}, \quad \tilde{q}_A^{\dot{a}} \equiv |p\rangle^{\dot{a}} \eta_A$

→ Satisfy the general supersymmetry Algebra

$$\{q^{Aa}, \tilde{q}_B^{\dot{b}}\} = -\delta_B^A p^{a\dot{b}}$$

proof:

$$\begin{aligned} & [P]^a \frac{\partial}{\partial \eta_A} (|p\rangle^{\dot{b}} \eta_B) + |p\rangle^{\dot{b}} \eta_B \left( [P]^a \frac{\partial}{\partial \eta_A} \right) \\ &= [P]^a |p\rangle^{\dot{b}} \delta_B^A - [P]^a |p\rangle^{\dot{b}} \eta_B \frac{\partial}{\partial \eta_A} + |p\rangle^{\dot{b}} \eta_B [P]^a \frac{\partial}{\partial \eta_A} \\ &= \delta_B^A |p\rangle^{\dot{b}} [P]^a = -\delta_A^B p^{a\dot{b}} \quad \text{as claimed.} \end{aligned}$$

\* Relationship between on-shell & off-shell superspace

\*  $N=1$  off-shell:  $\mathcal{M}^{(4|1)}$   $(x^\mu, \theta^a, \bar{\theta}^{\dot{a}})$

$$\{Q_a, \bar{Q}_{\dot{a}}\} = i(\sigma^\mu_{a\dot{a}}) \partial_\mu$$

$$\hookrightarrow Q_a = \frac{\partial}{\partial \theta^a}$$

$$\bar{Q}_{\dot{a}} = -\frac{\partial}{\partial \bar{\theta}^{\dot{a}}} + i(\theta^a)(\sigma^\mu)_{a\dot{a}} \partial_\mu$$

$$\{Q_a, \bar{Q}_{\dot{a}}\} = \left\{ \frac{\partial}{\partial \theta^a}, i\theta^b(\sigma^\mu)_{b\dot{a}} \partial_\mu \right\}$$

$$= i\delta_a^b \sigma^\mu_{b\dot{a}} \partial_\mu - i\theta^b(\sigma^\mu_{b\dot{a}}) \partial_\mu \frac{\partial}{\partial \theta^a} + i\theta^b \sigma^\mu_{b\dot{a}} \partial_\mu \frac{\partial}{\partial \theta^a}$$

$$= -i\sigma^\mu_{a\dot{a}} \partial_\mu = -\sigma^\mu_{a\dot{a}} P_\mu$$

• Can now go on-shell: i.e.  $P^\mu P_\mu = 0 \Rightarrow P_\mu \mapsto P_{a\dot{a}}$

$$\text{or } P_{a\dot{a}} = -[P]_a \langle P |_{\dot{a}}$$

$$\text{Thus: } \bar{Q}_{\dot{a}} = -\frac{\partial}{\partial \bar{\theta}^{\dot{a}}} = \theta^a [P]_a \langle P |_{\dot{a}}$$

$$\text{and so } \{Q_a, \bar{Q}_{\dot{a}}\} = [P]_a \langle P |_{\dot{a}}$$

\* Can introduce Grassmann variables  $\theta^a$

$$\mathcal{N} \equiv \theta^a [P]_a \quad \frac{\partial}{\partial \theta^a} = \frac{\partial}{\partial \theta^a} \mathcal{N} = [P]_a \frac{\partial}{\partial \mathcal{N}}$$

$$\{Q_a, \tilde{Q}^a\} = \left\{ \frac{\partial}{\partial \theta^a}, \theta^a [P]_a \langle P|_a \right\}$$

$$= \{ [P]_a \frac{\partial}{\partial \mathcal{N}}, \theta^a [P]_a \langle P|_a \}$$

$$= \{ [P]_a \frac{\partial}{\partial \mathcal{N}}, \mathcal{N} \langle P|_a \} = [P]_a \langle P|_a$$

Raising both indices:

$$\{Q^a, \tilde{Q}^a\} = \{ [P]^a \frac{\partial}{\partial \mathcal{N}}, \mathcal{N} |P\rangle^a \} = |P\rangle^a [P]^a$$

Thus, for  $N=4$ :  $\{Q^{Aa}, \tilde{Q}_B^b\} = \{ [P]^a \frac{\partial}{\partial \mathcal{N}}, \mathcal{N}_B |P\rangle^a \}$   
as claimed

\* The supercoords  $\theta$ : have mass-dim  $[\theta] \sim (\text{mass})^{1/2}$   
and  $[|P\rangle] \sim (\text{mass})^{1/2}$  so  $\mathcal{N}$  are dimensionless (as they should be).

$$\mathcal{N} = \theta^a [P]_a$$

So under little group rescaling  $\mathcal{N} \rightarrow t^{-1} \mathcal{N}$   
 $\Rightarrow$  On-shell chiral Superfield scales as

$$\Omega \rightarrow t^{-2} \Omega$$

SuperAmplitudes  $\Omega = \Omega(p_i)$  "Super wave functions"  
for the  $i^{\text{th}}$  external particle of  $a$  (color ordered)

Superamplitude  $\mathcal{A}_n[\Omega_1 \dots \Omega_n]$ ,  $(p_i, \mathcal{N}_i^A)$   $i=1 \dots n$

$\rightarrow$  we can expand  $\mathcal{A}_n[\Omega_1 \dots \Omega_n]$  for which we find it to  
be a sum of polynomials in  $\mathcal{N}_{iA}$  of degree  $4(a+2)$

\* Can extract states using the differential operators at  $n=0$

ex: (1)

$$\Delta_n[1^+ \dots i^- \dots j^- \dots n^+] = \left( \prod_{A=1}^4 \frac{\partial}{\partial \mathcal{N}_{iA}} \right) \left( \prod_{B=1}^4 \frac{\partial}{\partial \mathcal{N}_{jB}} \right) \mathcal{A}_n[\Omega_1 \dots \Omega_n] \Big|_{\mathcal{N}_{kC}=0}$$

(2)

$$\Delta_n[\bar{5}^{12} \bar{5}^{34} \bar{3}^{-4+} \dots n^+] = \left( \frac{\partial}{\partial \mathcal{N}_{11}} \frac{\partial}{\partial \mathcal{N}_{12}} \right) \left( \frac{\partial}{\partial \mathcal{N}_{23}} \frac{\partial}{\partial \mathcal{N}_{24}} \right) \left( \prod_{A=1}^4 \frac{\partial}{\partial \mathcal{N}_{3A}} \right) \mathcal{A}_n[\Omega_1 \dots \Omega_n] \Big|_{\mathcal{N}_{kC}=0}$$

(c)  $\frac{\partial}{\partial \mathcal{N}_{11}} \frac{\partial}{\partial \mathcal{N}_{12}} \mathcal{A}_n[\Omega_1 \dots \Omega_n]$

$$\sim \frac{1}{2!} \frac{\partial^2}{\partial \mathcal{N}_{11} \partial \mathcal{N}_{12}} (\mathcal{N}_A^1 \mathcal{N}_B^1 S^{AB}) = \frac{1}{2!} \delta_A^1 \delta_B^1 S^{AB} = \frac{1}{2!} \mathcal{N}_A \delta_B^1 S^{AB}$$

$$= \frac{1}{2!} \mathcal{N}_1^2 S^{1B} = \mathcal{N}_1 S^{11} = \mathcal{N}_1^1 S^{11}$$

$$\therefore \frac{\partial}{\partial \lambda_{12}} (\lambda_{12}^2 \delta^{12}) = \delta_{12}^2 \delta^{12} = \delta^{12}$$

\* The order  $k$  of the Grassmann polynomial precisely corresponds to the  $N^k$  MHV sector

$$A_n = A_n^{\text{MHV}} + A_n^{\text{NMHV}} + A_n^{\text{N}^2\text{MHV}} + \dots + A_n^{\text{Anti-MHV}}$$

MHV:  $2 \rightarrow (n-2)$  Scattering

$$g^- g^- \quad g^+ \dots g^+ \quad A_n[- - + \dots +]$$

NMHV:  $3 \rightarrow (n-3)$   $A_n[- - - + \dots +]$

$N^k$  MHV:  $(k+2) \rightarrow n-k-2$   $A_n$

Anti-MHV:  $(n-2) \rightarrow 2$   $A_n[1^- \dots i^+ \dots j^+ \dots n^-] = \frac{[ij]^4}{[12][23] \dots [n1]}$

$$- \quad g^- \quad g^+$$

$A_n^{\text{MHV}}$ : Grassmann Degree 8

$A_n^{\text{NMHV}}$ : " " 12

$$\rightarrow A^{\text{MHV}} = A_n[g^- g^- g^+ \dots g^+] (\eta_1)^4 (\eta_2)^4 + A_n[g^- \lambda^{123} \lambda^4 g^+ \dots g^+] (\eta_1)^4 (\eta_{21} \eta_{22} \eta_{23}) (\eta_{34})^4$$

$$\text{where } (\eta_i)^4 = \eta_{i1} \eta_{i2} \eta_{i3} \eta_{i4}$$

\* SWI in the Language of on-shell superspace:

$$Q^A A_n = 0$$

$$\tilde{Q}_A A_n = 0$$

$$Q^A \equiv \sum_{i=1}^n q_i^A = \sum_{i=1}^n [i | \frac{\partial}{\partial \eta_{iA}}$$

$$\tilde{Q}_A \equiv \sum_{i=1}^n \tilde{q}_{iA} = \sum_{i=1}^n |i\rangle \eta_{iA}$$

$$\eta_{iA} \equiv \theta_A^a [P_i]_a$$

$$A_n = A_0 \bigoplus_{k=1}^{n-4} A^k$$

\* To see that  $Q^A A_n = 0$  describes the SWI

we find that each sector is independent. So we

$$\text{can look at } Q^A A_0 = Q^A A^{\text{MHV}} = 0$$

$A^{\text{MHV}}$  has Grassmann Degree 8,

$$= A_n[g^- g^- g^+ \dots g^+] (\eta_1)^4 (\eta_2)^4 + A_n[g^- \lambda^{123} \lambda^4 \dots g^+] (\eta_1)^4$$

$$+ A_n[g^- s^{12} s^{34} g^+ \dots g^+] (\eta_1)^4 (\eta_2^{(2)} \eta_3^{(2)}) (\eta_1^{(3)} \eta_2^{(3)}) + \dots$$

$$A_n[g^+ \dots g_i^- \dots g_j^- \dots g^+] (\eta_i)^4 (\eta_j)^4 + \dots A_n[\lambda^{123} g^- \lambda^4 g^+ \dots g^+] (\eta_1^4 \eta_2^{(2)} \eta_3^{(2)}) (\eta_2^4 \eta_3^{(3)})$$

$\rightarrow$  Pick  $A = 4$

$$\begin{aligned}\tilde{Q}_A \mathcal{A}^{M+V} &= \sum_{i=1}^n |i\rangle n_i^{(1)} \mathcal{A}^{M+V} = 0 \\ &= |3\rangle A_n [g^- g^- g^+ - g^+ ] (n_1)^4 (n_2)^4 n_4^{(3)} + |2\rangle n_4^{(2)} (A_n [g^- \lambda^{123} g^+ ] ) \\ &\quad + |1\rangle n_4^{(1)} A_n [\lambda^{123} g^- - \lambda^4 g^+ \dots g^+ ] (n_1)^4 n_2^{(1)} n_3^{(1)} (n_2)^4 n_4^{(3)} + \dots \\ &= (|3\rangle A_n [g^- g^- g^+ \dots g^+ ] - |2\rangle A_n [g^- \lambda^{123} \lambda^4 g^+ - g^+ ] - |1\rangle A_n [\lambda^{123} g^- \lambda^4 g^+ ] ) \\ &\quad \times (n_1)^4 (n_2)^4 n_4^{(3)}\end{aligned}$$

$\rightarrow$  Recovers the SWI from earlier.

\* The requirement  $\{Q^A, \tilde{Q}_B\} = \sum_i p_i^M$  annihilates the superamplitude (formally) is equivalent to the statement of momentum conservation

$$\{Q^A, \tilde{Q}_B\} \mathcal{A} = 0$$

$\rightarrow$  The associated delta-fn  $\delta^4(\sum_{i=1}^n p_i)$  has been implicit.  $P^M = \sum_i p_i^M$

$$P^M \mathcal{A} = 0$$

$\rightarrow$  Acts multiplicatively and the vanishing result follows from the fact the fact that  $x\delta(x)$  has support nowhere

• Define  $\delta^{(8)}(\tilde{Q})$  Grassmann delta

$$\delta^{(8)}(\tilde{Q}) = \frac{1}{2^4} \prod_{A=1}^4 \tilde{Q}_{A\dot{\alpha}} \tilde{Q}_A^{\dot{\alpha}} = \frac{1}{2^4} \prod_{A=1}^4 \sum_{i,j=1}^n \langle ij \rangle^4 n_{iA} n_{jA}$$

$$\begin{aligned}\tilde{Q}_{A\dot{\alpha}} \tilde{Q}_A^{\dot{\alpha}} &= \left( \sum_i \langle i |_{\dot{\alpha}} n_{iA} \right) \left( \sum_j |j\rangle^{\dot{\alpha}} n_{jA} \right) \\ &= \sum_{i,j} \langle ij \rangle n_{iA} n_{jA}\end{aligned}$$

Claim: Momentum conservation  $\Rightarrow Q^A \delta^{(8)}(\tilde{Q}) = 0$

$$\begin{aligned}\text{Proof: } Q^A \delta^{(8)}(\tilde{Q}) &= \left( \sum_K [K] \frac{\partial}{\partial n_{KA}} \right) \left( \frac{1}{2^4} \prod_{B=1}^4 \sum_{i,j=1}^n \langle ij \rangle n_{iB} n_{jB} \right) \\ &= \frac{1}{2^4} \sum_{i,j,k=1}^n \prod_{B=1}^4 \langle ij \rangle \left( \delta_i^K \delta_B^A n_{jB} - n_{iB} \delta_j^K \delta_B^A \right) [K] \\ &= \frac{1}{2^4} \left( \sum_{j,k} \prod_B \langle kj \rangle n_{jB} [k] - \sum_{i,k} \prod_B n_{iB} \langle ik \rangle [k] \right) \\ &= \frac{1}{2^4} \sum_{i,k} \prod_B n_{iB} \langle ki \rangle [k] \\ &= -\frac{1}{2^4} \sum_i \prod_B n_{iB} \left( \sum_k [k] \right)\end{aligned}$$

$$= \sum_{i \in B} \sum_{j \in B} \langle i j \rangle \langle j i \rangle \dots = 0 \text{ via momentum conservation}$$

\* Also have  $\tilde{Q}_A \delta^{(8)}(\tilde{Q}) = 0$

- Half of the supersymmetry constraints ( $\tilde{Q}_A \mathcal{A}_n = 0$ ) are satisfied if we write  $\mathcal{A}$  as an exact "form"

$$A_n^{MHV} = \delta^{(8)}(\tilde{Q}) P_n^{(4K)}$$

$P_n^{(4K)}$ : degree  $4K$  polynomial in  $\mathcal{N}$

- If  $Q^4 P = 0 \quad \forall A$ , then all the susy constraints are solved

•  $\delta^{(8)}$  is also written as  $\delta^{4|4}$

$$\delta^{4|4} = \prod_{a=1}^4 \bar{\delta}(z^a) \prod_{\dot{a}=1}^4 \chi^{\dot{a}} = \Lambda^3 \bar{\partial} \left( \frac{1}{z^a} \right) \prod_{a=1}^4 \chi^a$$

$(0,4)$  on  $\mathbb{C}^{4|4}$

$\Rightarrow \delta^{(8)}(\tilde{Q})$ : Conservation of (super) momenta

- $\delta^{(8)}(\tilde{Q})$  is of degree 8 so  $\delta^{(8)}(\tilde{Q})$  fixes MHV amplitudes completely and  $P_n^{(4K)}$  is a normalization constant

$$N=4 \text{ SYM (tree level)}: A_n^{MHV}[123\dots n] = \frac{\delta^{(8)}(\tilde{Q})}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$$

$\Rightarrow$  extracting Parke-Taylor

$$A_n[i^+ \dots i^- \dots j^- \dots j^+ \dots n^+] = (\partial_i)^4 (\partial_j)^4 A_n^{MHV} \Big|_{\eta_{KB}=0}$$

$$(\partial_i)^4 \delta^{(8)}(\tilde{Q}) = (\partial_i)^4 \prod_{k \in B} \langle k l \rangle \eta_{kB} \eta_{lB}$$

$$\begin{aligned} & \left( \sum_{k \in B} \langle k l \rangle (\partial_i)^4 (\eta_1^{(k)} \eta_1^{(l)} \eta_2^{(k)} \eta_2^{(l)} \dots \eta_u^{(k)} \eta_u^{(l)}) \right) \\ &= \left( \sum_{k \in B} (\partial_1^{(i)} \partial_2^{(i)} \partial_3^{(i)} \partial_4^{(i)} (\eta_1^{(k)} \eta_1^{(l)} \dots \eta_u^{(k)} \eta_u^{(l)})) \right) \\ &= \left( \sum_{k \in B} \partial_1^{(i)} \partial_2^{(i)} \partial_3^{(i)} \left[ \eta_1^{(k)} \eta_2^{(l)} \dots (\delta^{kl} \eta_u^{(l)} - \eta_u^{(k)} \delta^{le}) \right] \right) \\ &= \left( \sum_{k \in B} \left[ (\delta^{ik} \eta_1^{(l)} - \eta_1^{(k)} \delta^{il}) (\delta^{ik} \eta_2^{(l)} - \eta_2^{(k)} \delta^{le}) \right. \right. \\ & \quad \left. \left. \dots (\delta^{ik} \eta_u^{(l)} - \eta_u^{(k)} \delta^{le}) \right] \right) \\ &= \left( \sum_{k \in B} \langle i l \rangle \eta_1^{(l)} \eta_2^{(l)} \eta_3^{(l)} \eta_u^{(l)} + \dots \right) \end{aligned}$$

Now apply  $(\partial_j)^4$ :

$$\Rightarrow (\partial_i)^4 [ ] = \left( \sum_{k \in B} (\delta^{ik} \delta^{lj} - \delta^{ki} \delta^{jl}) (\delta^{ik} \delta^{lj} - \delta^{kl} \delta^{ji}) \right)$$

$$= \frac{1}{2^4} \sum_{kl} (\delta^{ik} \delta^{lj} - \delta^{kj} \delta^{il})^4 \langle kl \rangle^4$$

$$= \langle ij \rangle^4$$

$$\therefore A_n [1^+ \dots i^- \dots j^- \dots n^+] = (\partial_i)^4 (\partial_j)^4 A_n^{MHV} \Big|_{\eta_{kl}=0} = \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$$

\* Can reproduce word identities:  $A_n [g^- \lambda^{123} \lambda^4 g^+] = \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n [g^- g^- \dots g^+]$

$$\begin{aligned} A_n [g^- \lambda^{123} \lambda^4 g^+] &= (\partial_1)^4 (\partial_2^{(2)} \partial_3^{(2)} \partial_4^{(2)}) \partial_4^3 A_n^{MHV} \Big|_{\eta=0} \\ &= (\partial_1^{(2)} \partial_2^{(2)} \partial_3^{(2)}) (\partial_4^3) \left( \sum (\delta^{1k} \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{2l}) \dots \right) \\ &= (\partial_1^{(2)} \partial_2^{(2)} \partial_3^{(2)}) \sum (\delta^{1k} \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{2l}) \dots (\delta^{3k} \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{2l}) \\ &= \partial_1^{(2)} \partial_2^{(2)} \sum (\partial_1^4 \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{1k}) \dots (\delta^{2k} \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{2l}) \dots \\ &= \partial_1^{(2)} \sum (\dots) \dots (\delta^{2k} \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{2l}) \dots \\ &= \sum (\delta^{1k} \delta^{2l} - \delta^{k2} \delta^{1l}) (\delta^{2k} \eta^{(2)}_{kl} - \eta^{(2)}_{kl} \delta^{2l}) \dots \\ &\quad (\delta^{1k} \delta^{2l} - \delta^{2k} \delta^{1l}) \\ &= \langle 12 \rangle \langle 12 \rangle \langle 12 \rangle \langle 13 \rangle \end{aligned}$$

$$A_n [g^- \lambda^{123} \lambda^4 g^+] = \frac{\langle 12 \rangle^3 \langle 13 \rangle}{\langle 12 \rangle \dots \langle n1 \rangle} = \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n [g^- g^- \dots g^+]$$

as claimed:

\* The Word identities reproduces contractions a-la Wick

ex: above

$$A_n [g^- \lambda^{123} \lambda^4 g^+] = (\partial_1^4 \partial_2^{(1)} \partial_3^{(1)} \partial_4^{(1)}) (\partial_1^{(2)} \partial_2^{(2)} \partial_3^{(2)}) \partial_4^{(3)} A_n^{MHV}$$

$\langle 12 \rangle \langle 12 \rangle \langle 12 \rangle \langle 13 \rangle$

$$\begin{aligned} \text{(ii)} \quad A_n [g^- s^{12} s^{34} g^+ \dots g^+] &= (\partial_1)^4 (\partial_2^{(2)} \partial_3^{(2)}) (\partial_3^{(3)} \partial_4^{(3)}) A_n^{MHV} \\ &= \langle 12 \rangle \langle 12 \rangle \langle 13 \rangle \langle 13 \rangle \\ &\quad \langle 12 \rangle \dots \langle n1 \rangle \end{aligned}$$

$$= (\langle 13 \rangle^2 / \langle 12 \rangle^2) A_n [g^- g^- \dots g^+]$$

(iii) In general, we can prove  $A_n [g^+ \dots g_i^- \dots g_j^- \dots] = \frac{\langle ij \rangle^4}{\langle 12 \rangle \dots \langle n1 \rangle} A_n [-]$

Proof:  $A_n [g^+ \dots g_i^- \dots g_j^- \dots] = (\partial_i)^4 (\partial_j)^4 A_n^{MHV}$

$$= \frac{\langle ij \rangle^4}{\langle 12 \rangle \dots \langle n1 \rangle} = \frac{\langle ij \rangle^4}{\langle 12 \rangle^4} A_n$$

as claimed

Scalar scattering in  $N=4$ :

$$A_4[S^{12}S^{34}S^{12}S^{34}] = (\partial_1^{\mu_1}\partial_2^{\mu_1})(\partial_3^{\mu_2}\partial_4^{\mu_2})(\partial_1^{\mu_3}\partial_2^{\mu_3})(\partial_3^{\mu_4}\partial_4^{\mu_4})A_n^{\text{MHV}} \\ = \frac{\langle 13 \rangle \langle 13 \rangle \langle 24 \rangle \langle 24 \rangle}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$$

$$A_4[S^{12}S^{23}S^{34}S^{41}] = \frac{\langle 41 \rangle \langle 12 \rangle \langle 23 \rangle \langle 34 \rangle}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} = +1$$

$S^{41} = -S^{14}$

• Must "order" to use Wick

~> Anti-MHV 3pt:  $A_n^{\text{anti-MHV}}$   $k = -1 \sim \text{degree } 4$

$$A_3^{\text{anti-MHV}} = \frac{1}{[12][23][31]} S^{(4)} \left( [12]\eta_3 + [23]\eta_1 + [31]\eta_2 \right)$$

where

$$S^{(4)}(*) = \prod_{A=1}^4 ([12]\eta_{3A} + [23]\eta_{1A} + [31]\eta_{2A})$$

~> Can also show that  $Q^A A_n^{\text{anti-MHV}} = \tilde{Q}_A A_n^{\text{anti-MHV}} = 0$

proof:  $\tilde{Q}_A A_n^{\text{anti-MHV}} = \sum_i |i\rangle \eta_{iA} A_n^{\text{anti-MHV}}$

$$= \sum_{i=1}^3 \frac{|i\rangle \eta_{iA}}{[12][23][31]} \prod_{A=1}^4 ([12]\eta_{3A} + [23]\eta_{1A} + [31]\eta_{2A})$$

$$= \sum_{i=1}^3 \frac{|i\rangle \eta_{iA}}{[12][23][31]} \prod_{A=1}^4 ([12]\eta_{3A} + [23]\eta_{1A} + [31]\eta_{2A})$$

• 3 part kinematics:  $[21] \propto [11]$

$$\left( \sum_{i=1}^3 |i\rangle [i1] \right) = 0 \quad \text{via momentum conservation}$$

Proof (i)  $Q^A A_n^{\text{anti-MHV}} = \left( \sum_i [i1] \frac{\partial}{\partial \eta_{iA}} \right) A_3^{\text{anti-MHV}}$

$$= \frac{\sum_i [i1] \frac{\partial}{\partial \eta_{iA}}}{[12][23][31]} \left( \prod_{B=1}^4 ([12]\eta_{3B} + [23]\eta_{1B} + [31]\eta_{2B}) \right)$$

$$= \left( \sum_i \frac{[i1]}{[12][23][31]} \right) ([12] \delta_{i3} \delta_B^A + [23] \delta_{i1} \delta_B^A + [31] \delta_{i2} \delta_B^A)$$

$$= \frac{1}{[12][23][31]} ([12][31] + [23][11] + [31][21])$$

$$= 0 \quad (\text{Schouten identity})$$